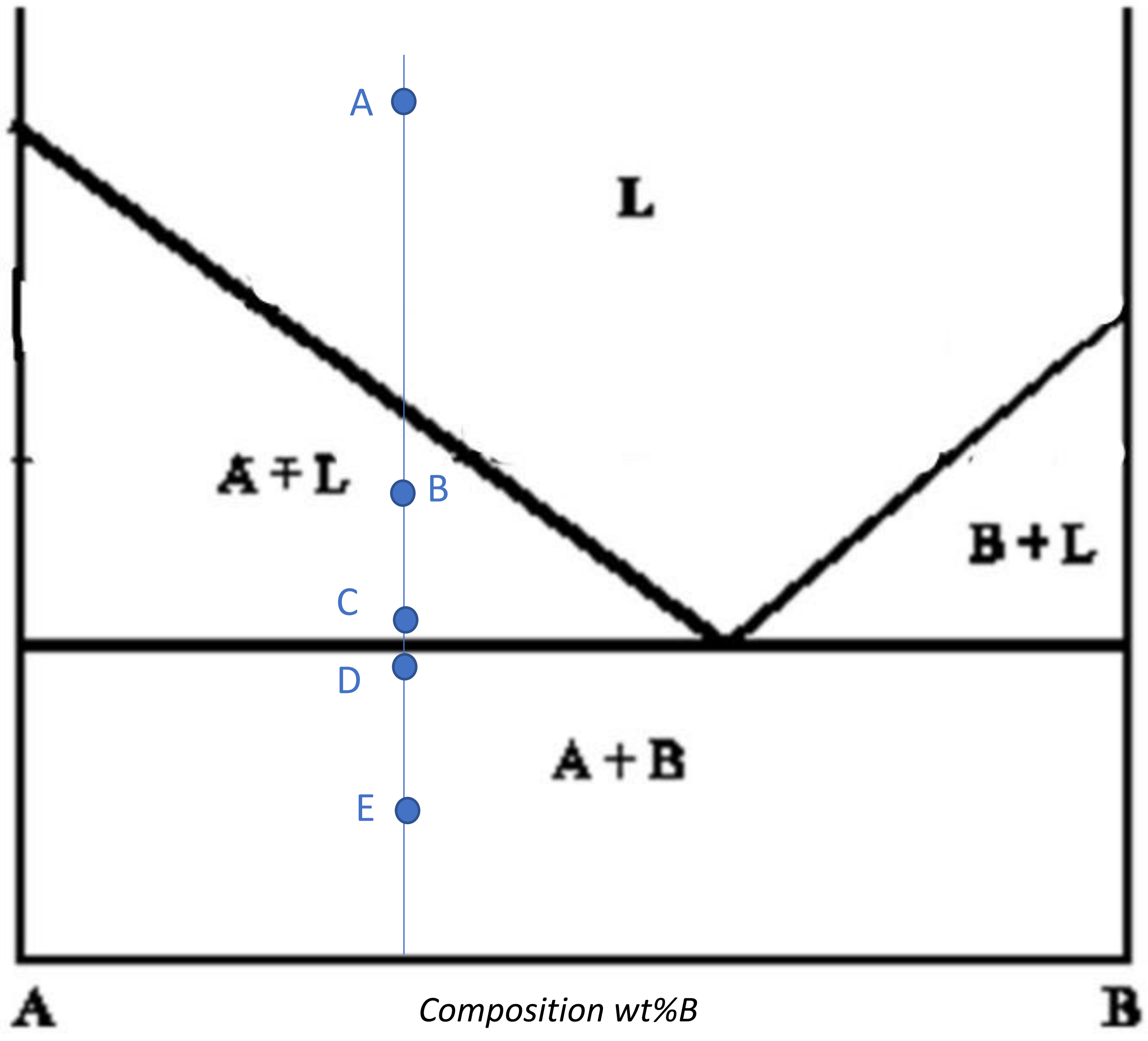
STM Exercise 6

Phase Diagrams

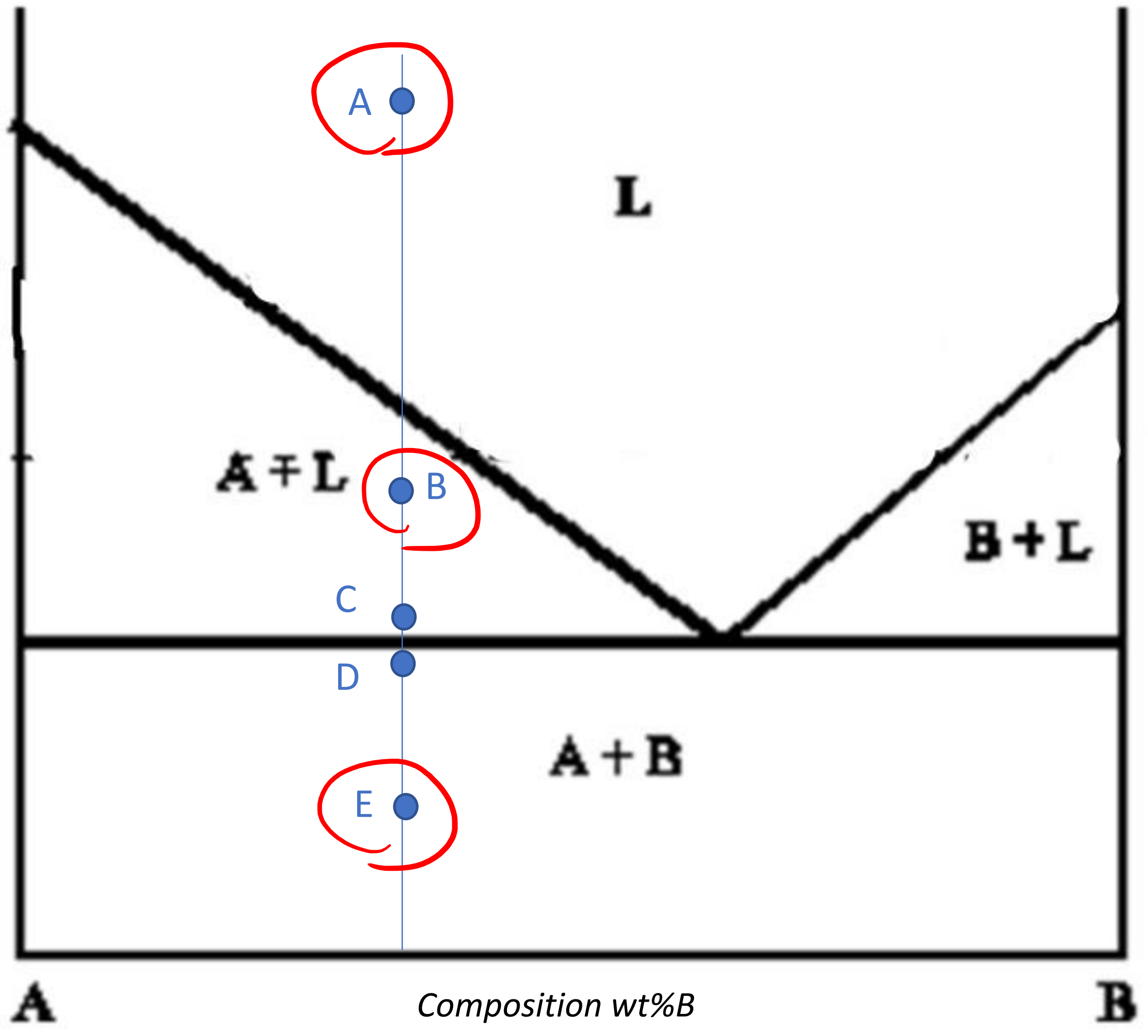
(no solubility in the solid state)

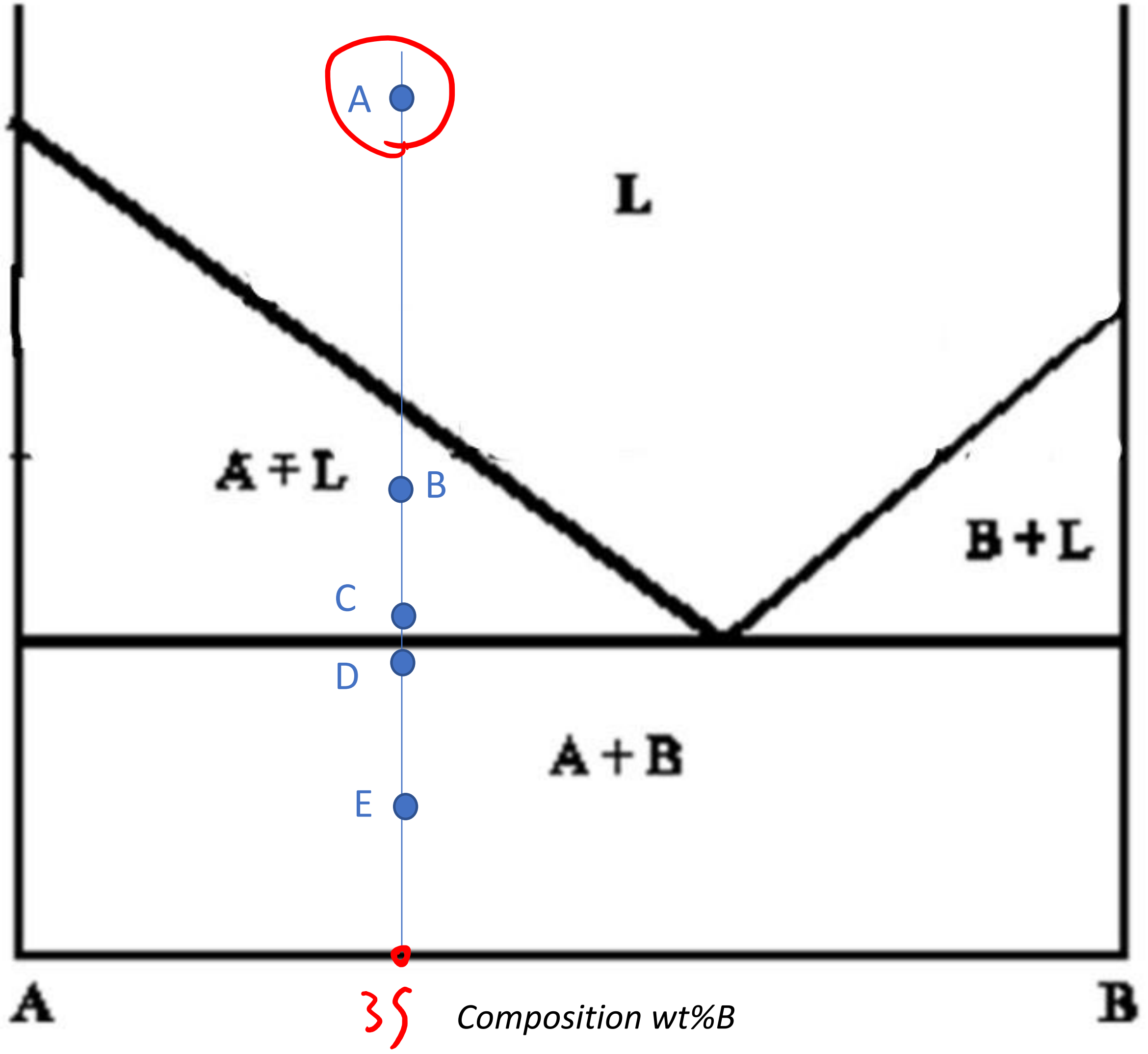
(partial solubility in the solid state)

Exercise 1: determine the phase present, their composition and relative amounts in the point A, B, C (at the beginning of the eutectic transformation), D (at the end of the eutectic transformation) and E indicated in the phase diagram. Schematize the microstructure in each point.



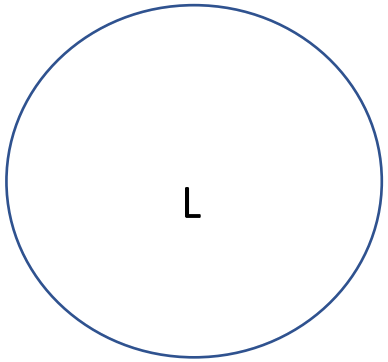
Exercise 1: determine the phase present, their composition and relative amounts in the point A, B, C (at the beginning of the eutectic transformation), D (at the end of the eutectic transformation) and E indicated in the phase diagram. Schematize the microstructure in each point.

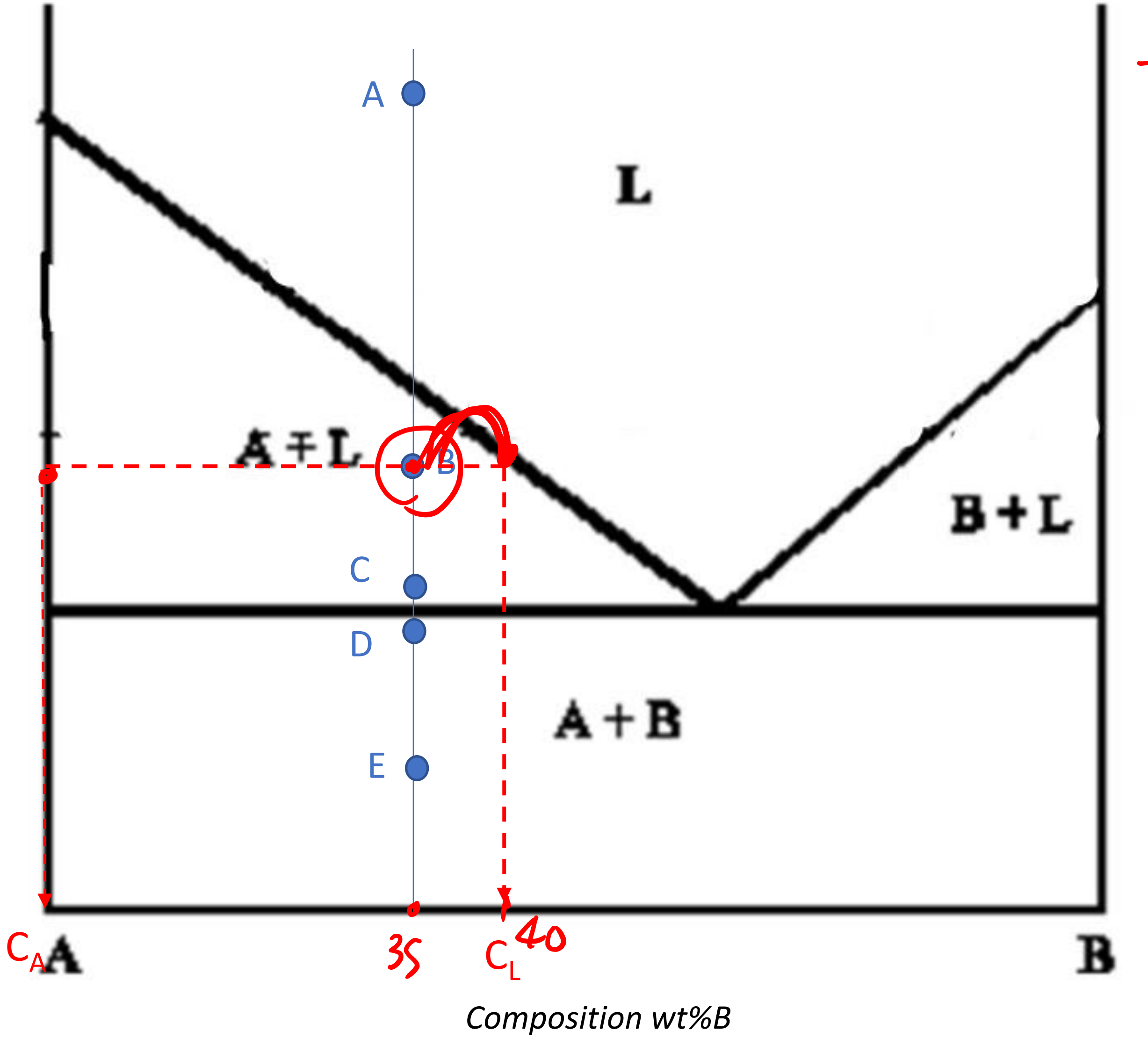




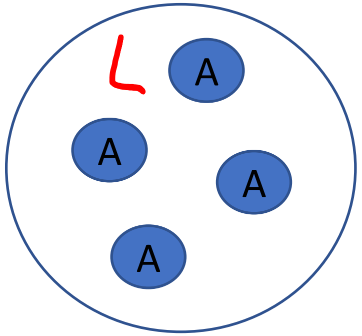
L

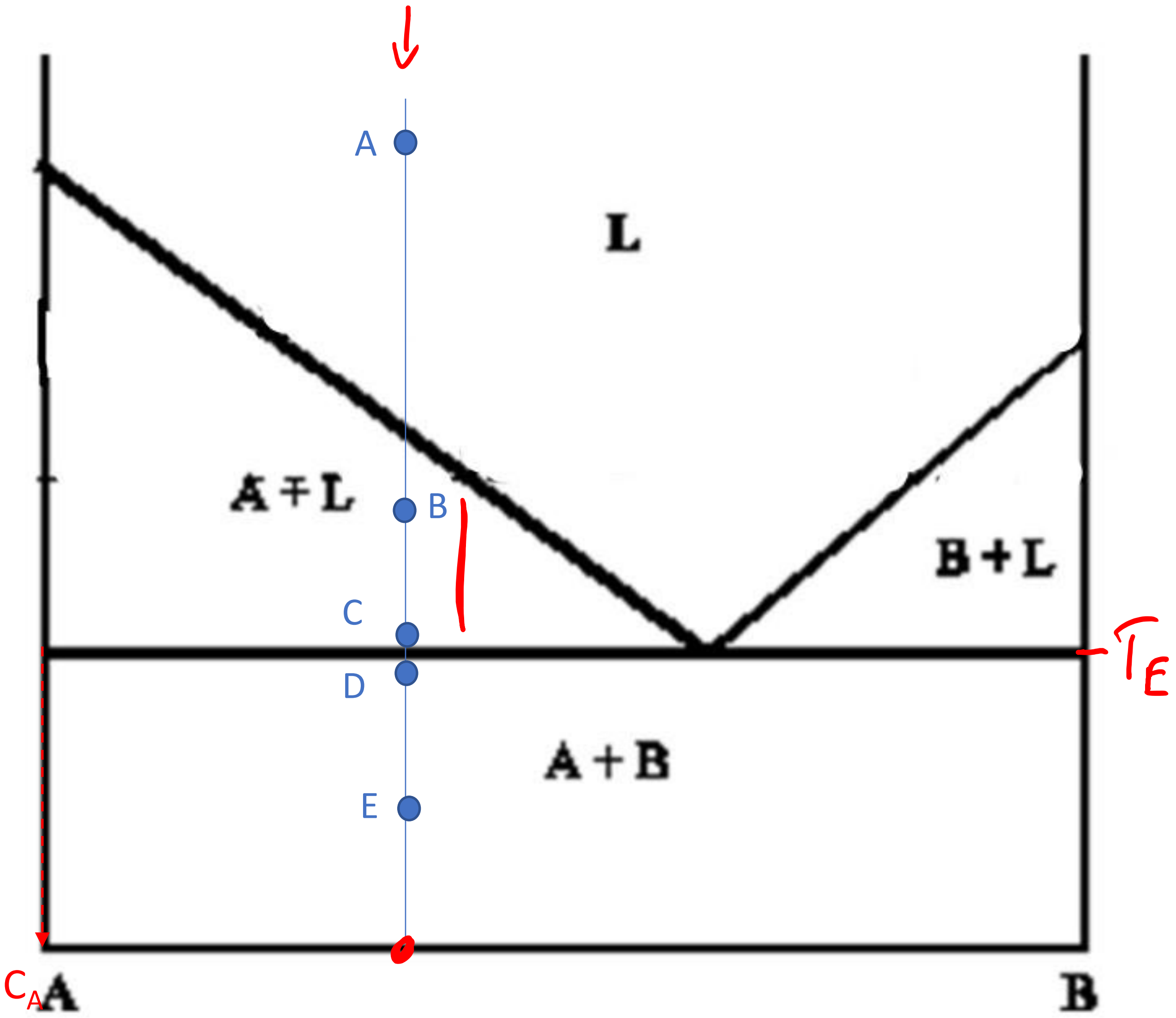
$C_L \begin{cases} 35\%\ B \\ 65\%\ A \end{cases}$



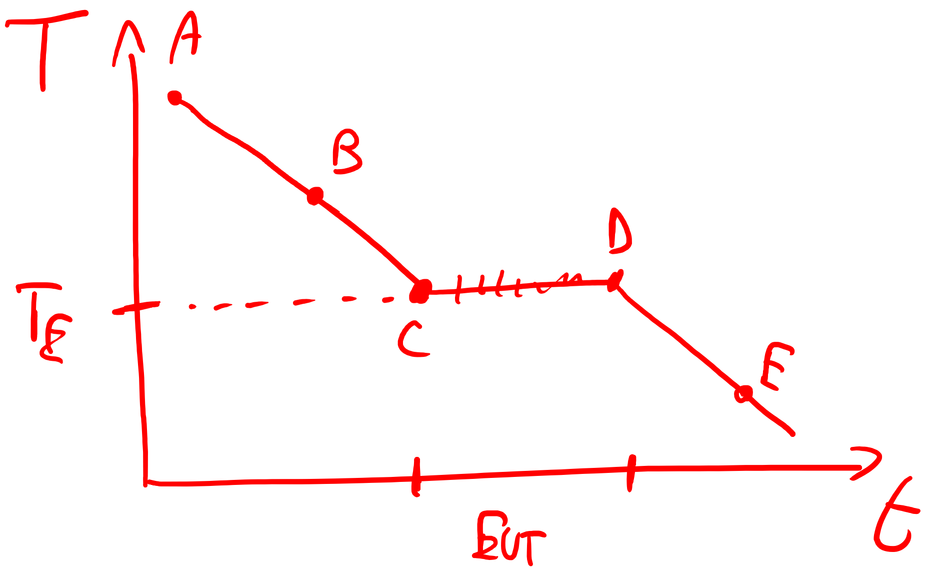


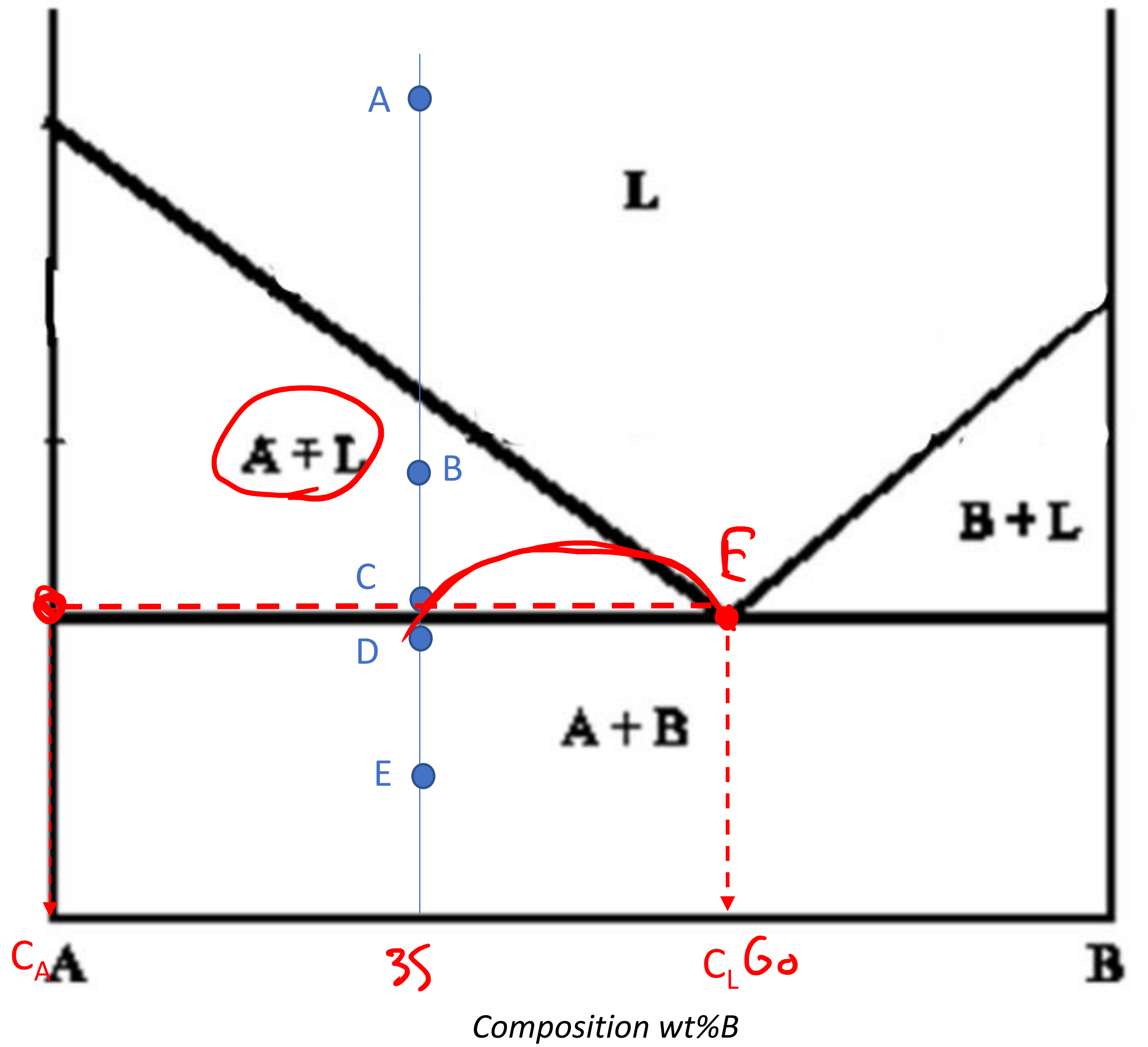
B	C	%
A	100% A	$\frac{40-35}{40}\cdot 100 = 12{,}5\%$
L	40% B 60% A	87,5%



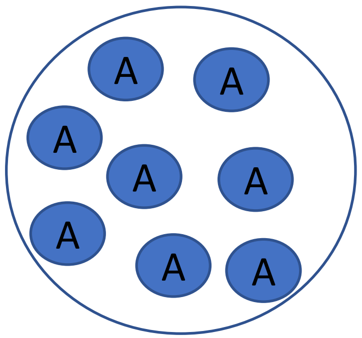


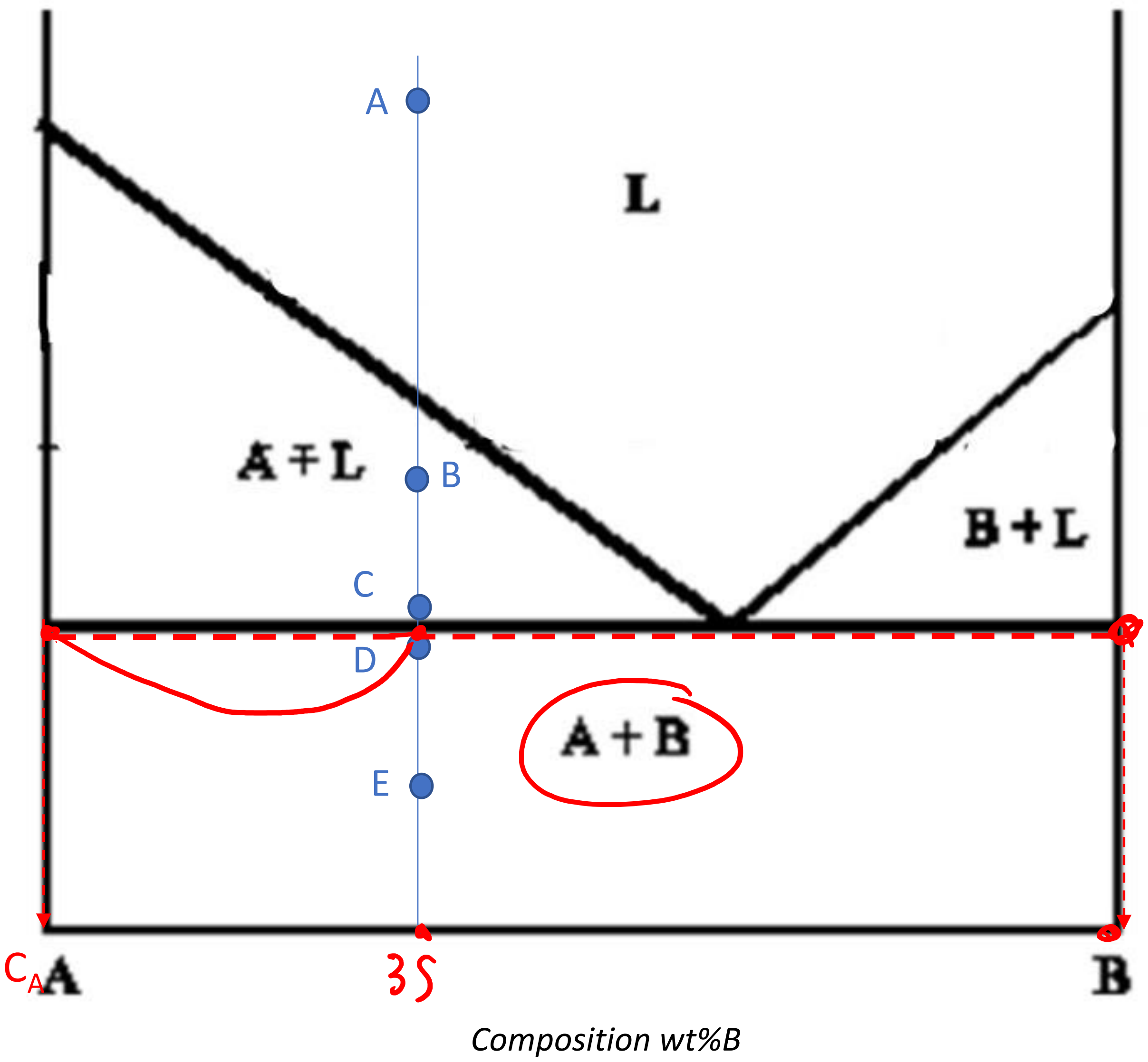
Composition wt%B





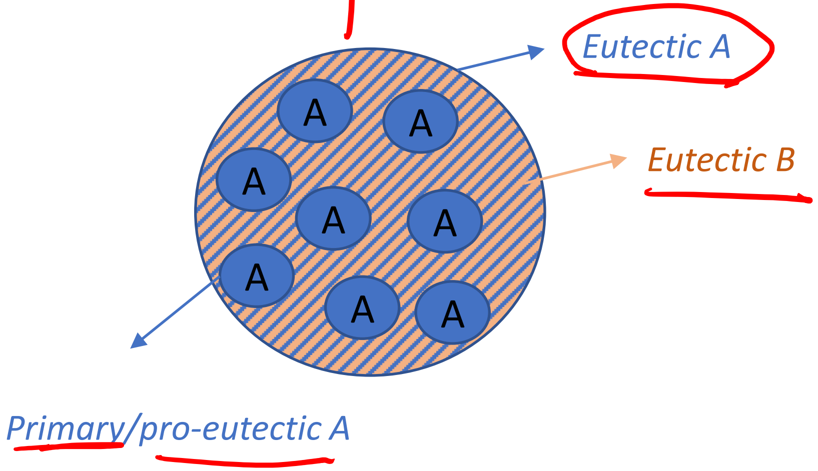
C	C	%
A	100%	$\frac{60-35}{60} \cdot 100 = 42\%$
L	60% B 40% A	58%

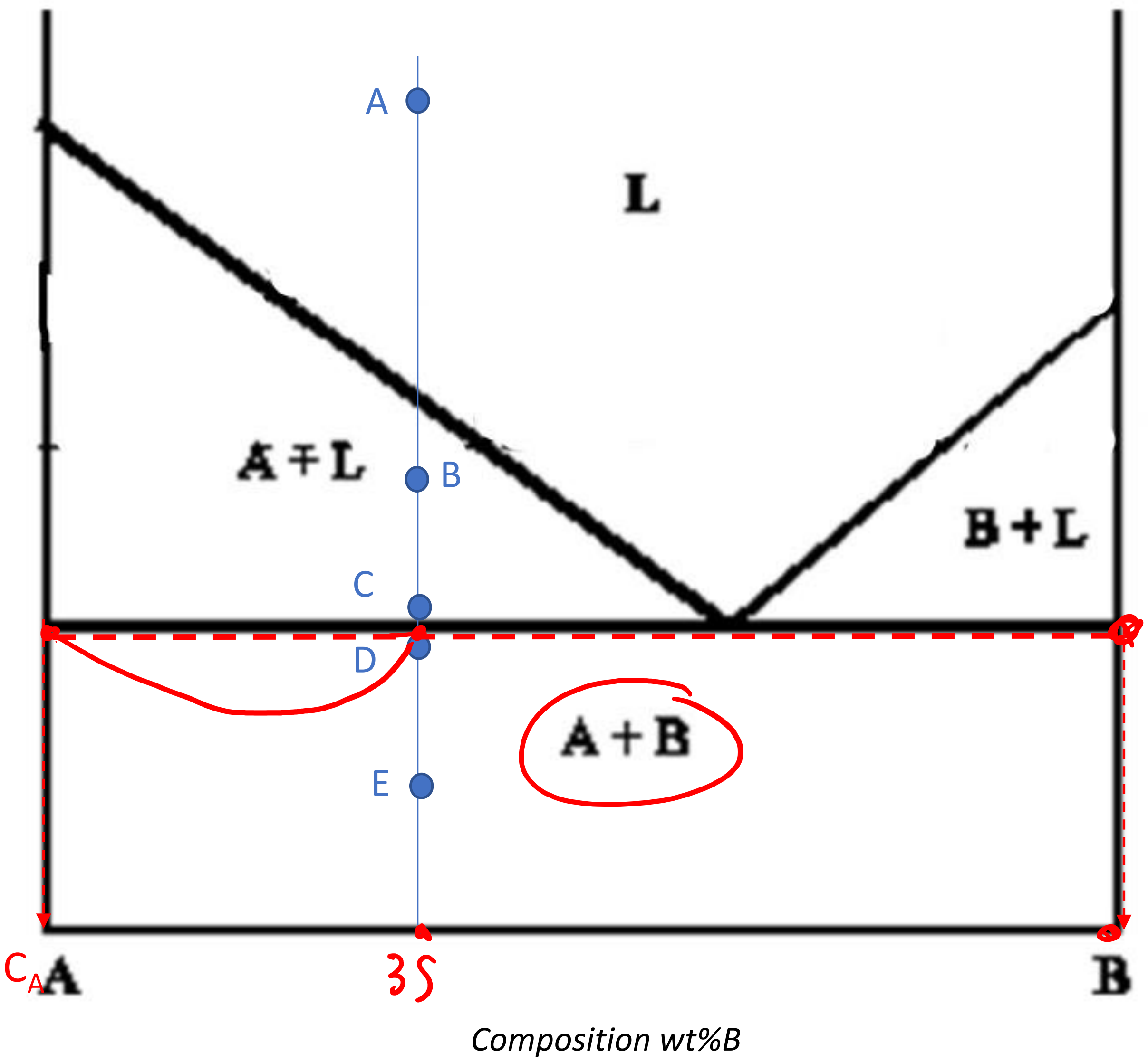




Ph → A + B

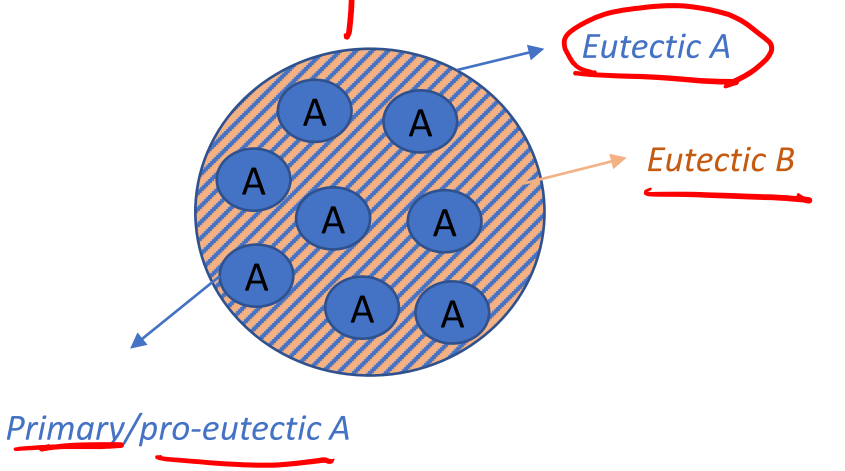
D E	C	%
A	100% A	65%
B	100% B	$\rightarrow \frac{35-0}{100-0}\cdot 100 = 35\%$





Ph → A + B

D E	C	%
A	100% A	65%
B	100% B	$\frac{35-0}{100-0} \cdot 100 = 35\%$

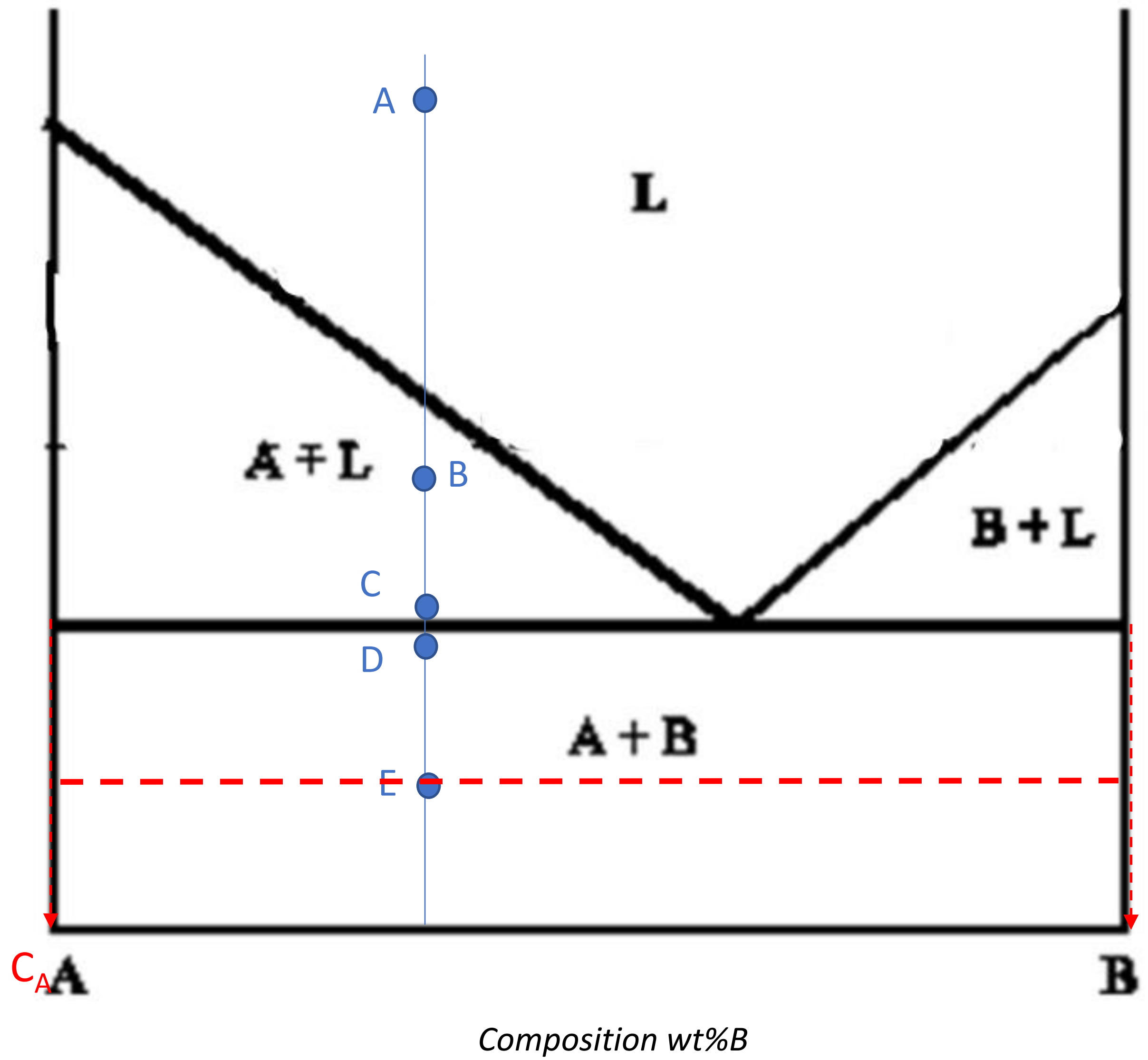


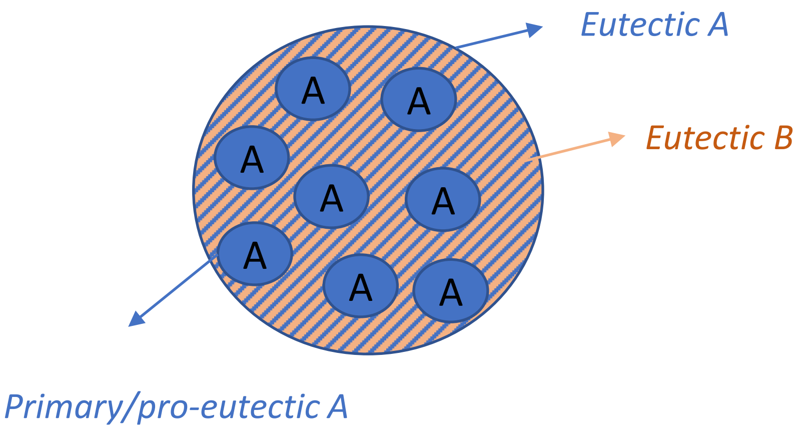
Primary/pro-eutectic A (As calculated in C, no more formed later):

% A= (60-35)/(60-0)*100=42%

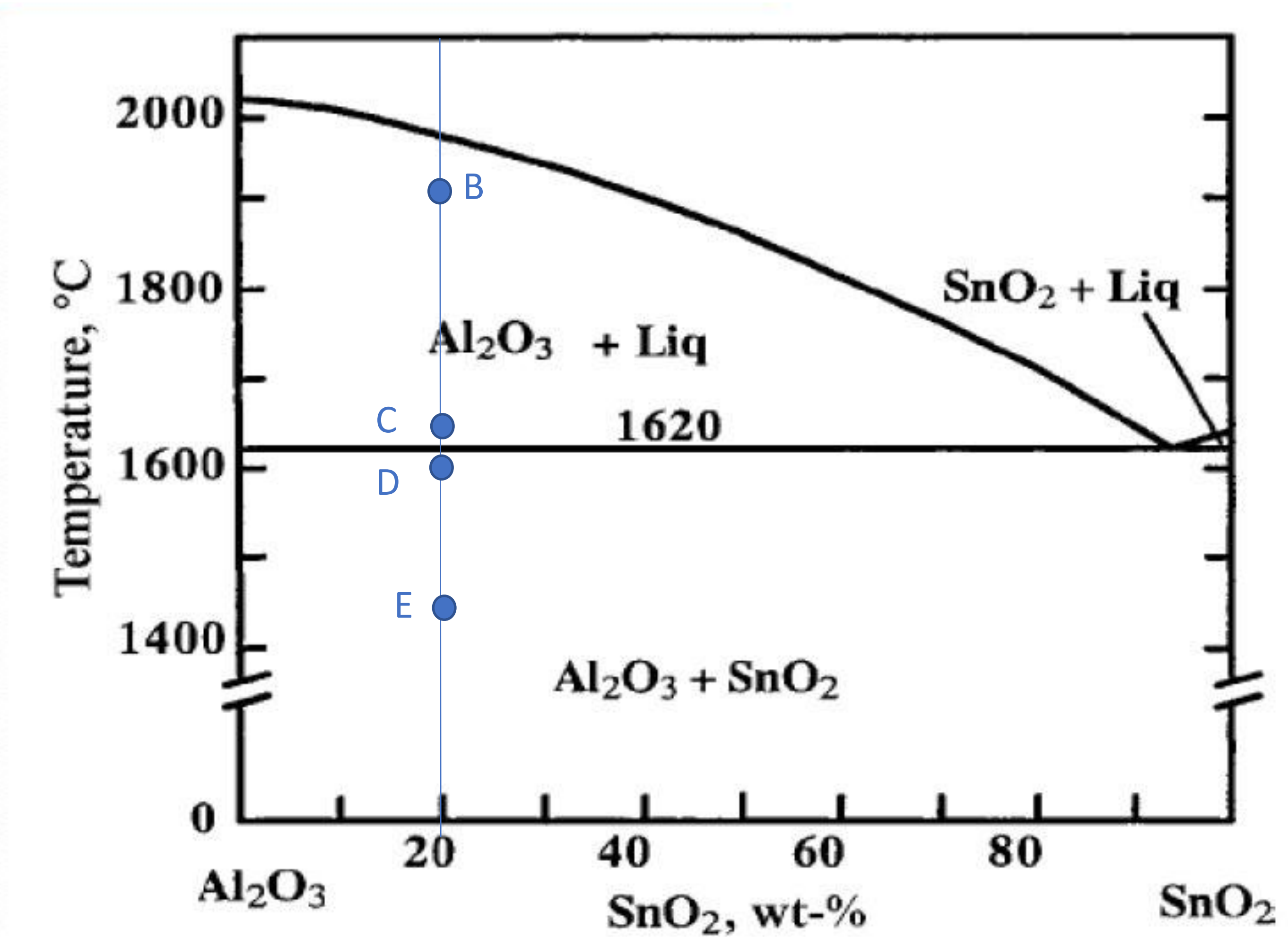
$\%A_{EUT}$= 65-42=23%

$E - D$


A
L
A+L
B
B+L
C
D
A+B
E
C_A A
B
Composition wt%B


Eutectic A
Eutectic B
A
Primary/pro-eutectic A

Exercise 2: determine the phase present, their composition and relative amounts in the points B, C (at the beginning of the eutectic transformation), D (at the end of the eutectic transformation) and E indicated in the phase diagram. Schematize the microstructure in each point.



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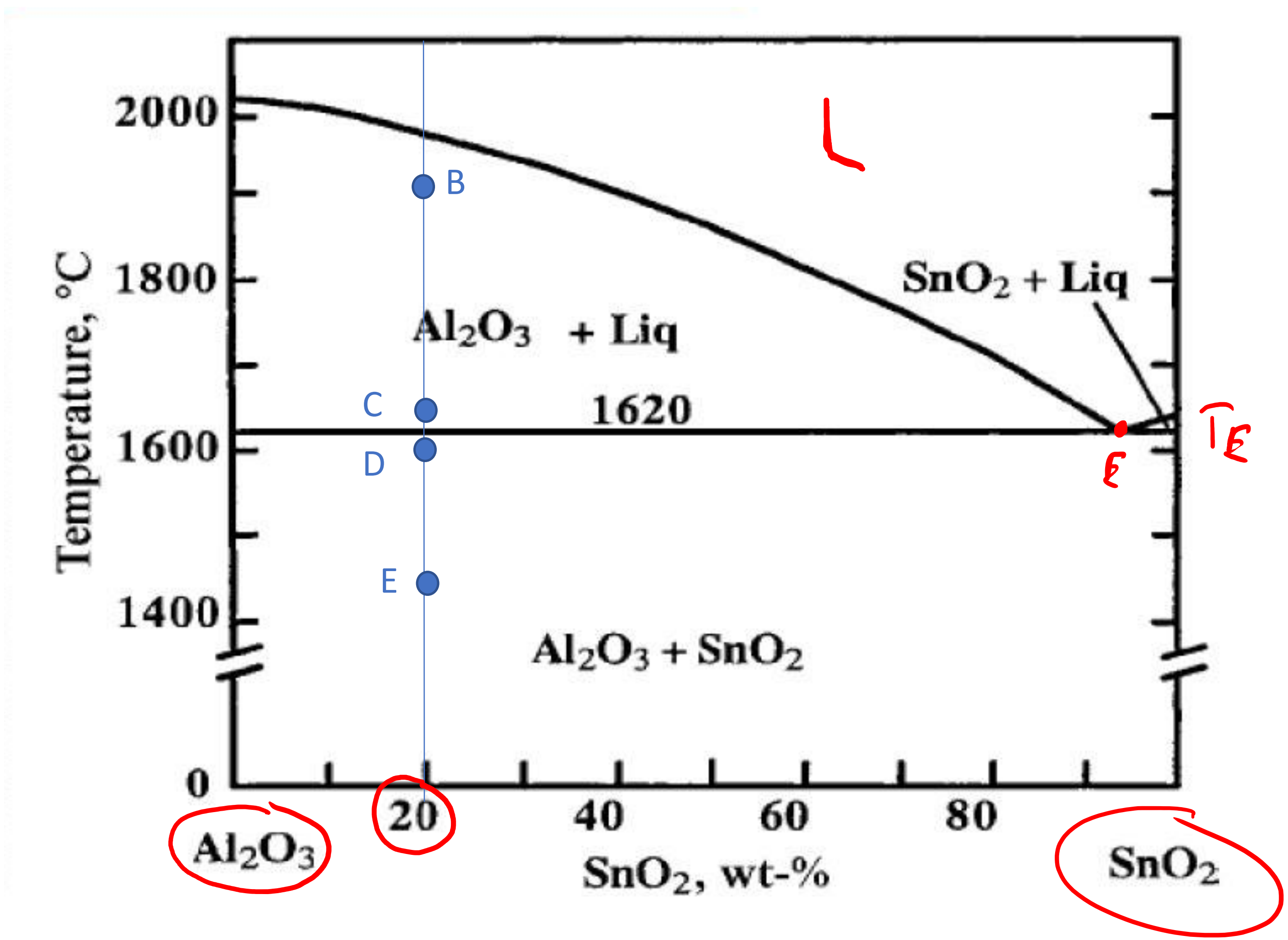

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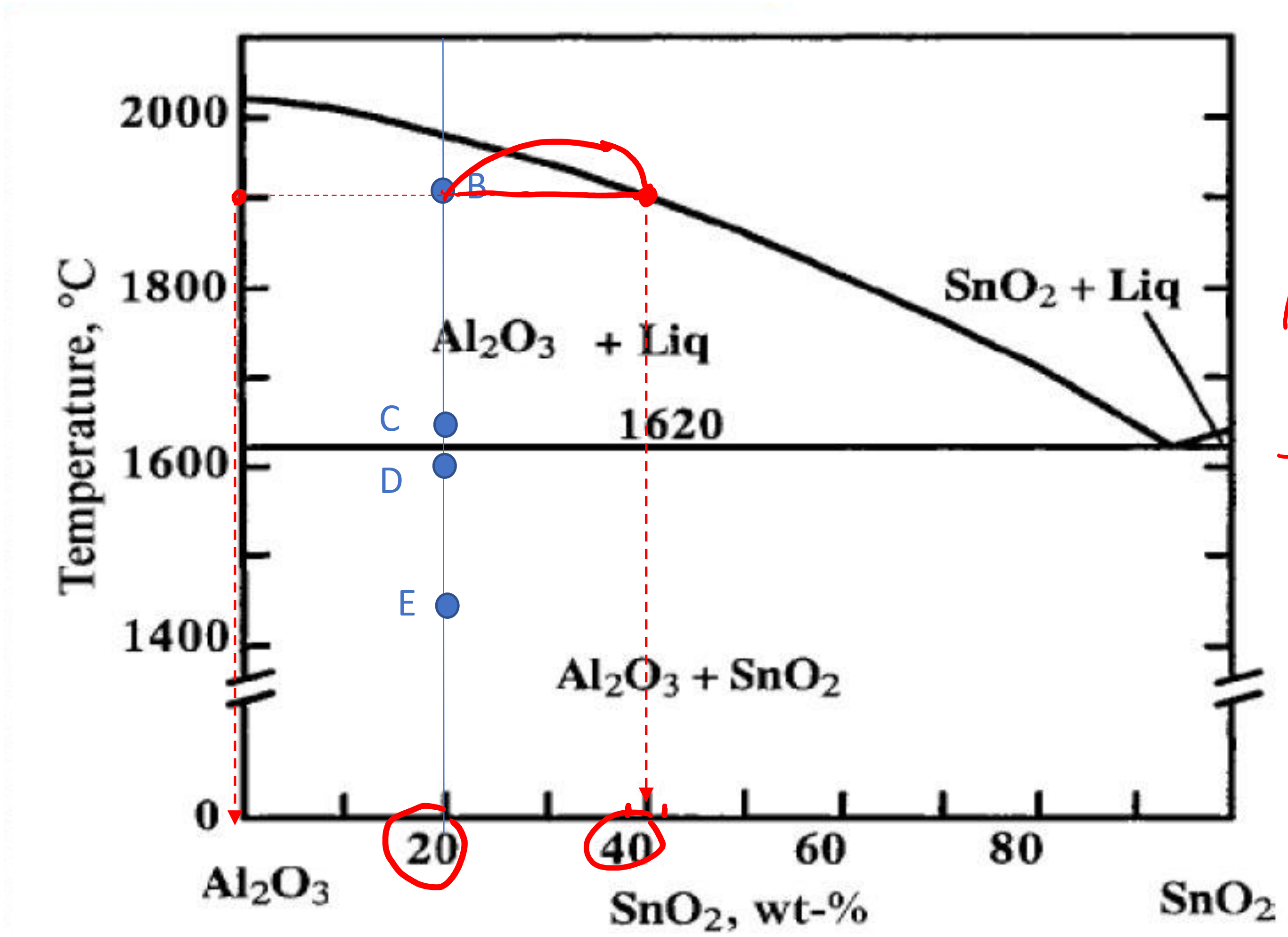
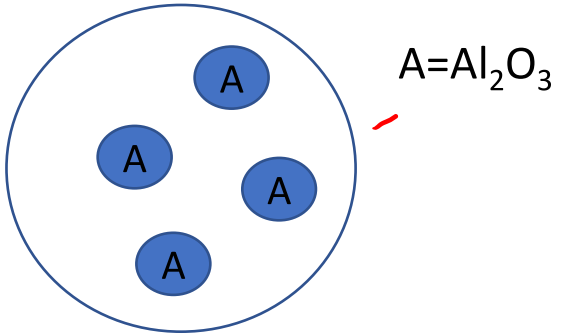


B	C	%
Al_2O_3	100% Al_2O_3	$\frac{40-20}{40} \cdot 100 = 50\%$
L	40% SnO_2 60% Al_2O_3	50%



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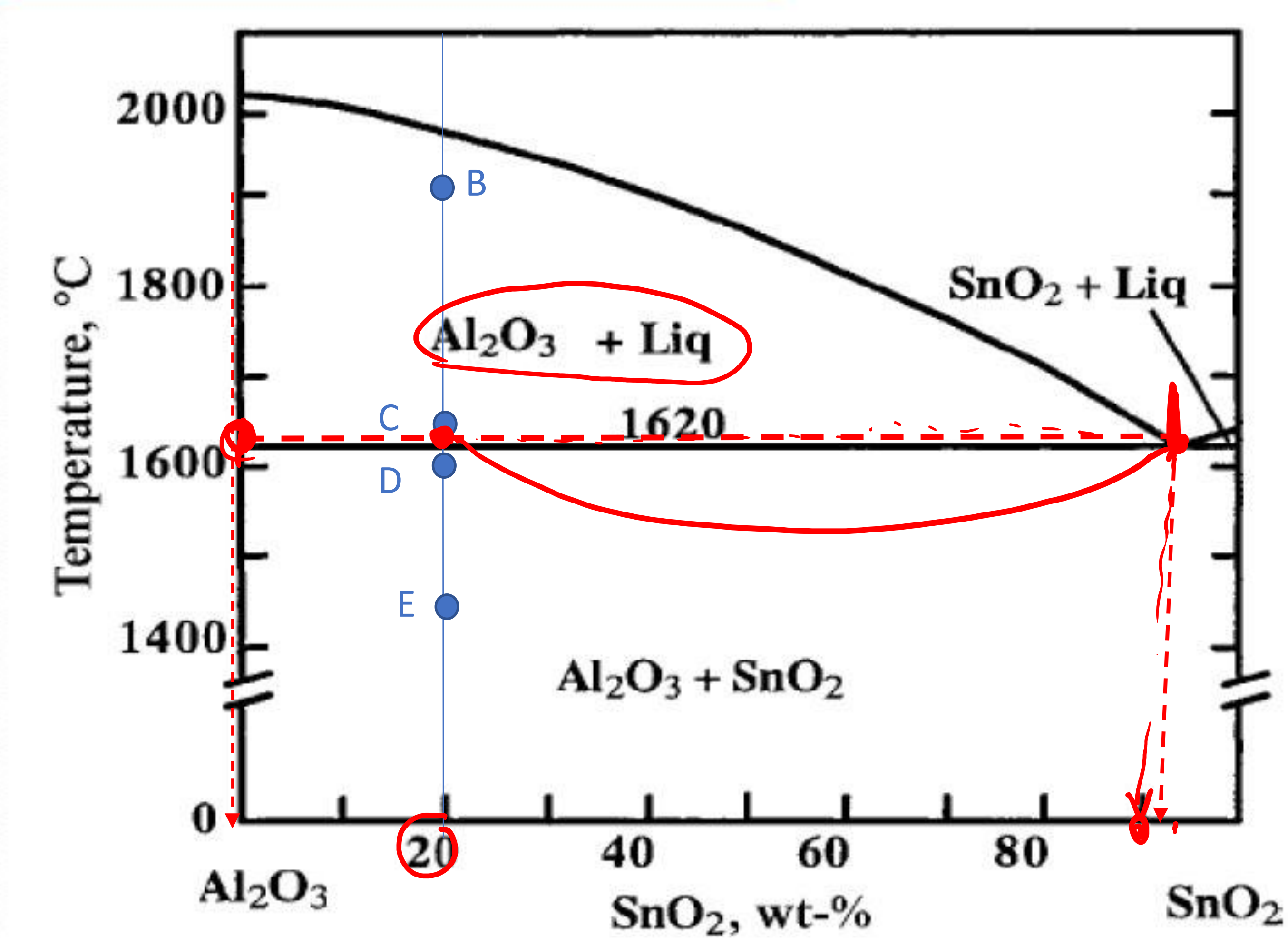
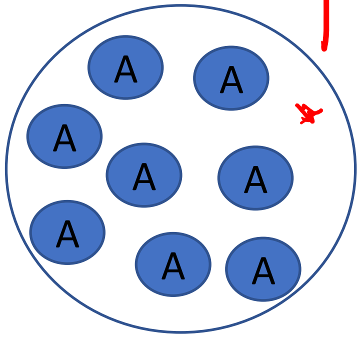
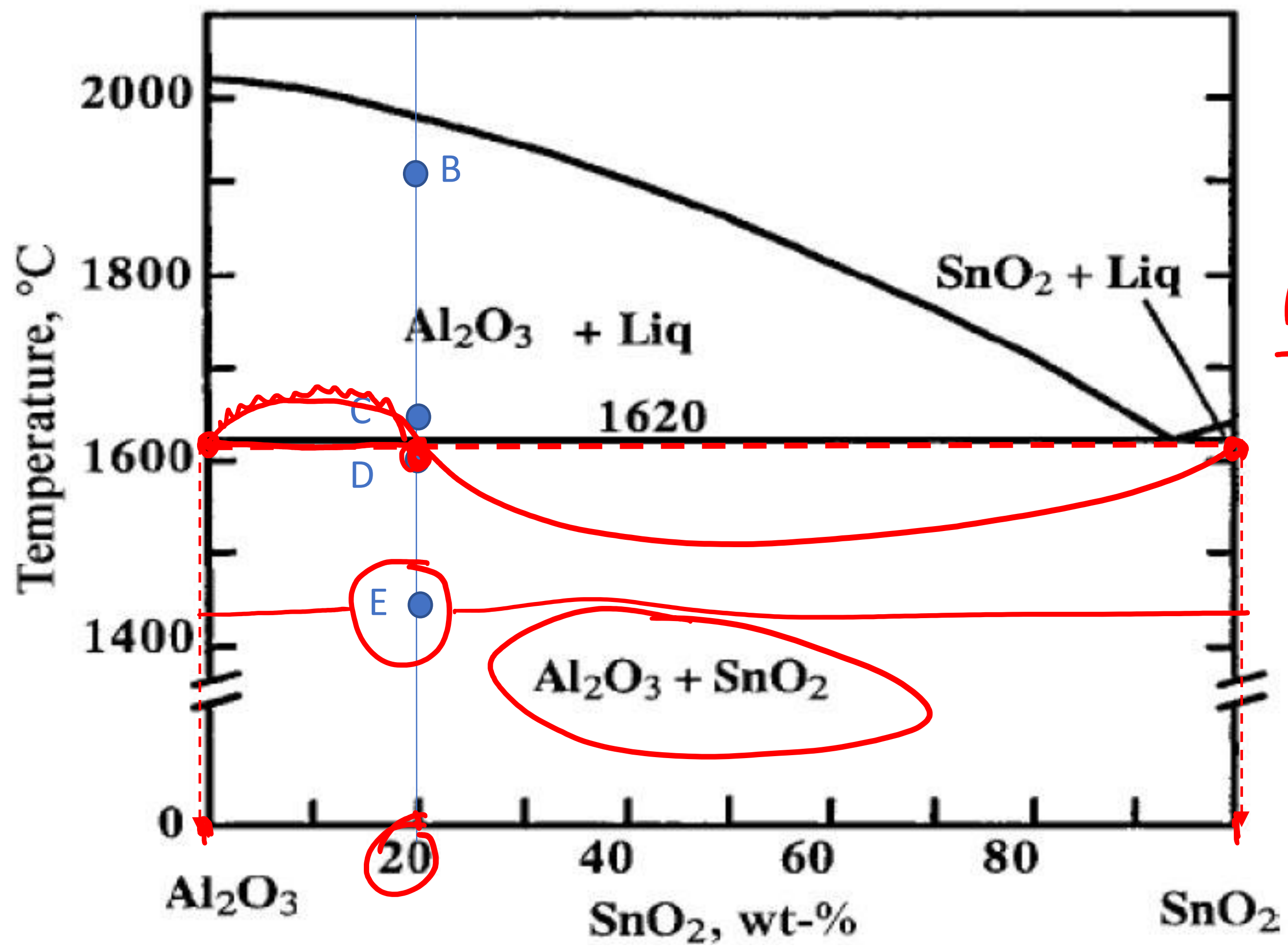


C	C	%
Al_2O_3	100% Al_2O_3	$\frac{90-20}{90} \cdot 100 = 78\%$
L	90% SnO_2 10% Al_2O_3	22%

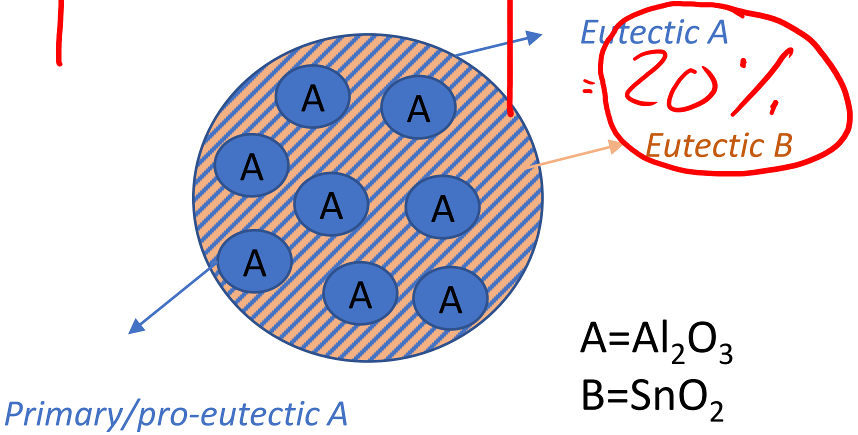


A=Al_2O_3

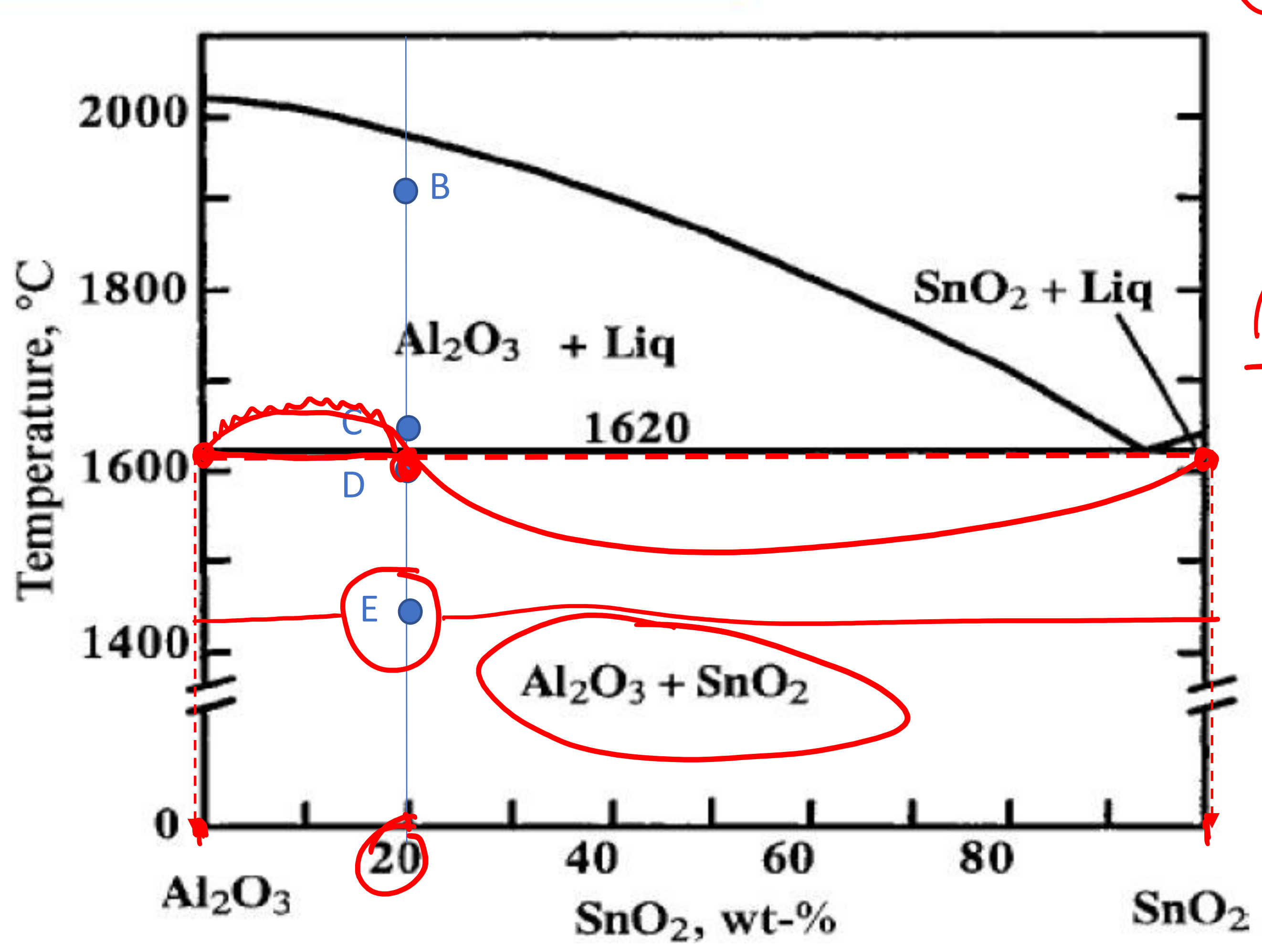


E''

D	C	%
Al_2O_3	100% Al_2O_3	80%
SnO_2	100% SnO_2	$\frac{20-0}{100-0} \cdot 100 =$ 20%

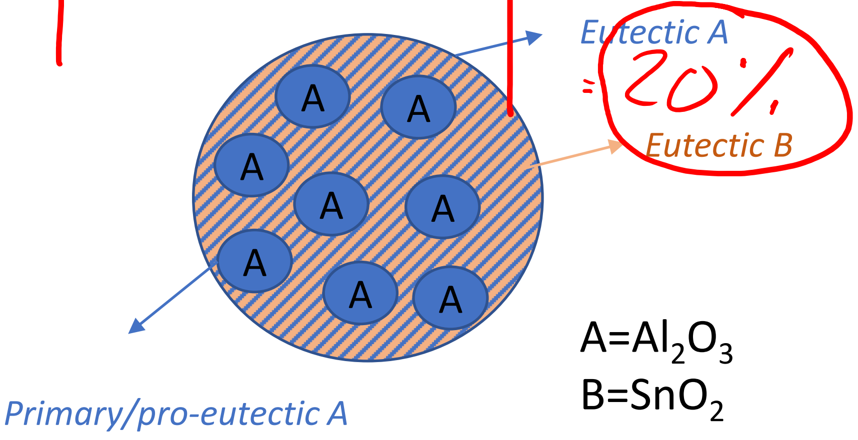


$A=Al_2O_3$
$B=SnO_2$



D	C	%
Al_2O_3	100% Al_2O_3	80%
SnO_2	100% SnO_2	$\frac{20-0}{100-0} \cdot 100 = 20\%$

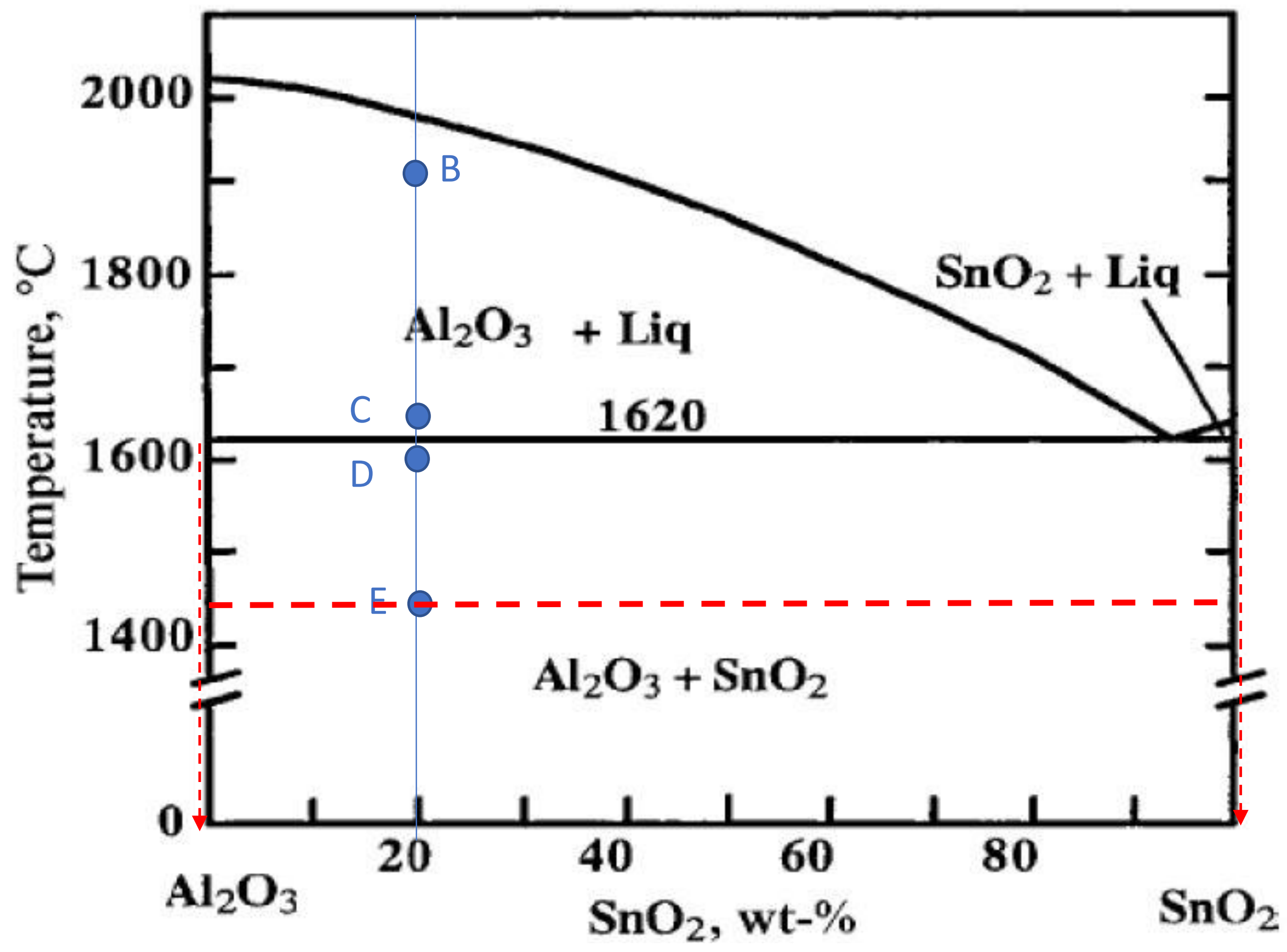
E''

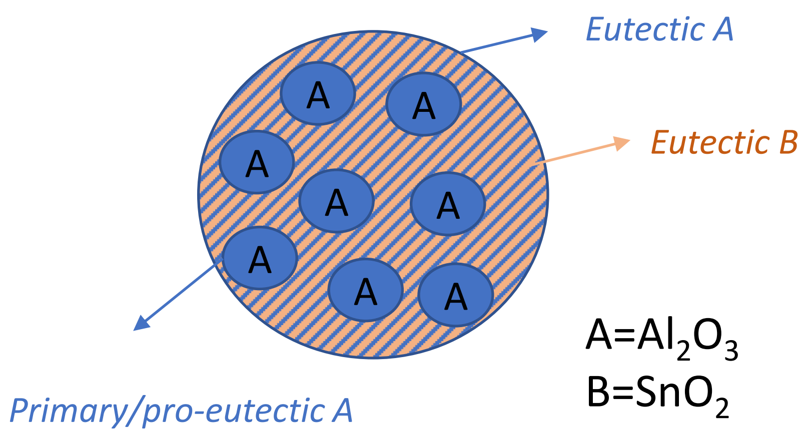


$A = Al_2O_3$
$B = SnO_2$

Primary/pro-eutectic Al_2O_3 (As calculated in C, no more formed later): **% Al_2O_3 $_{PRIM}$**= (92-20)/(92-0)*100=78%

$\%A_{EUT}$= 80-78=2%

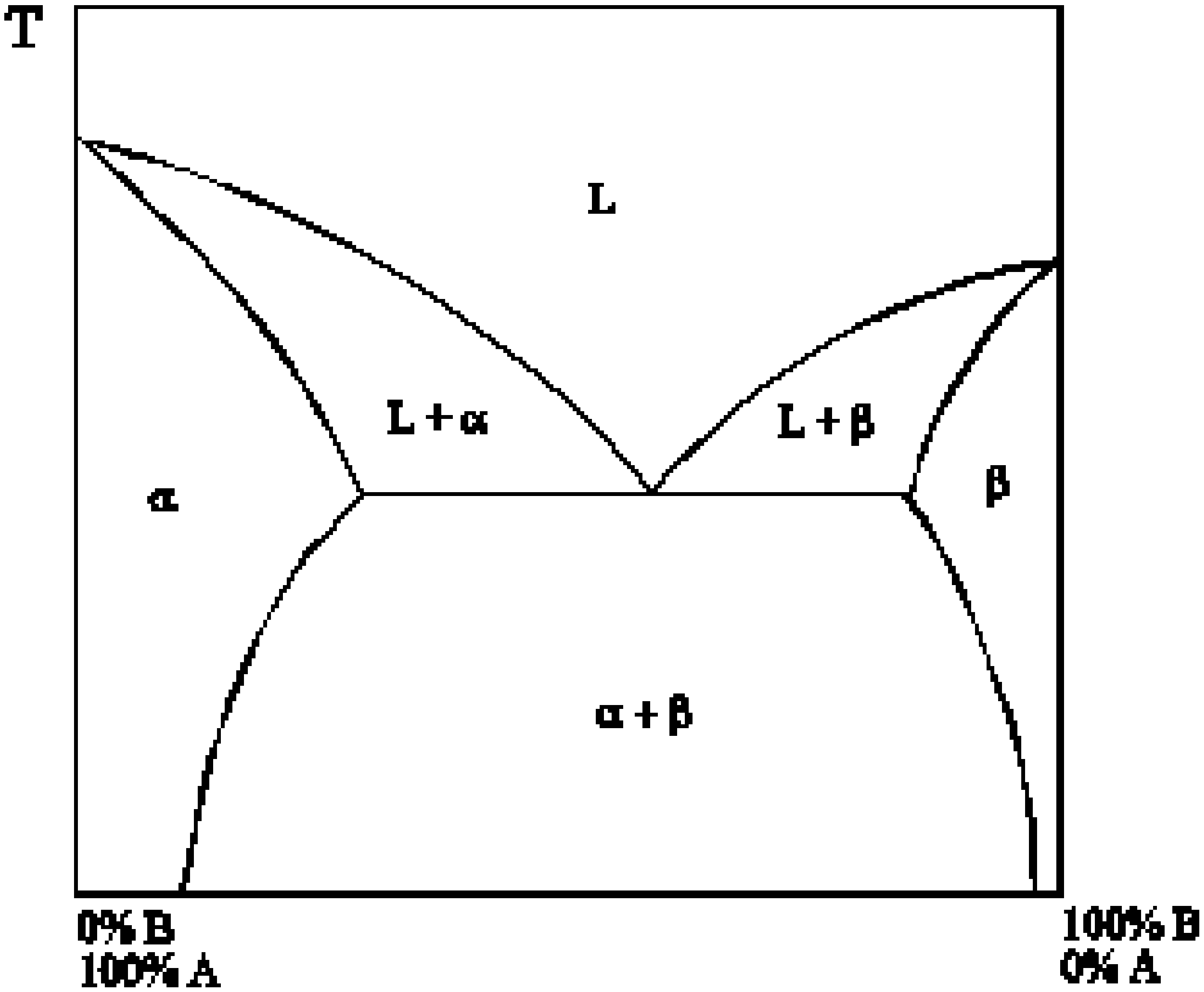




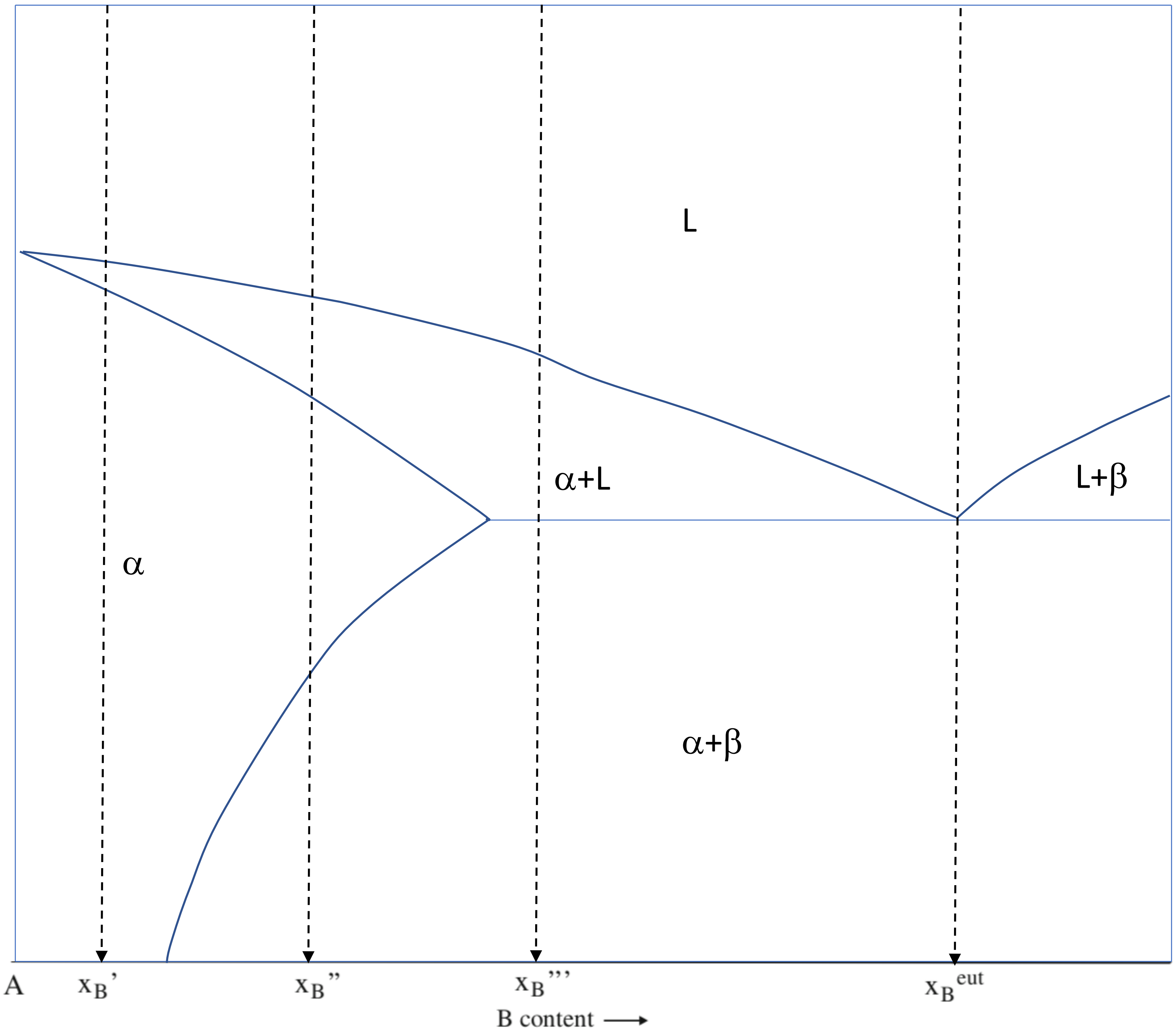
$A=Al_2O_3$
$B=SnO_2$

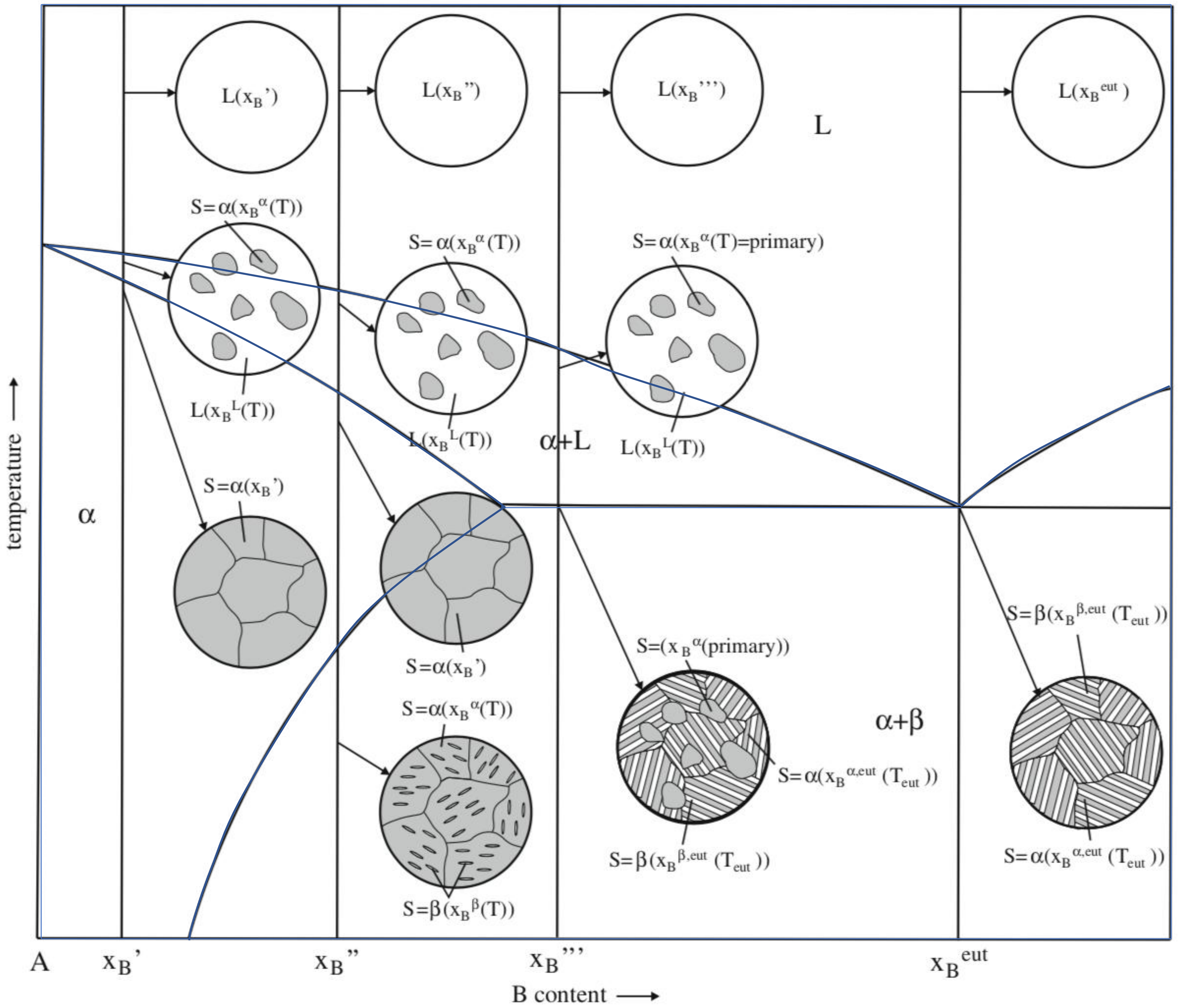
Exercise 3: Draw a phase diagram (for generic components A and B), with complete solubility in the liquid state and **partial solubility in the solid state**. Also indicate the phase/phases present(s) in each region of the phase diagram

Complete solubility in the liquid state, partial solubility in the solid state



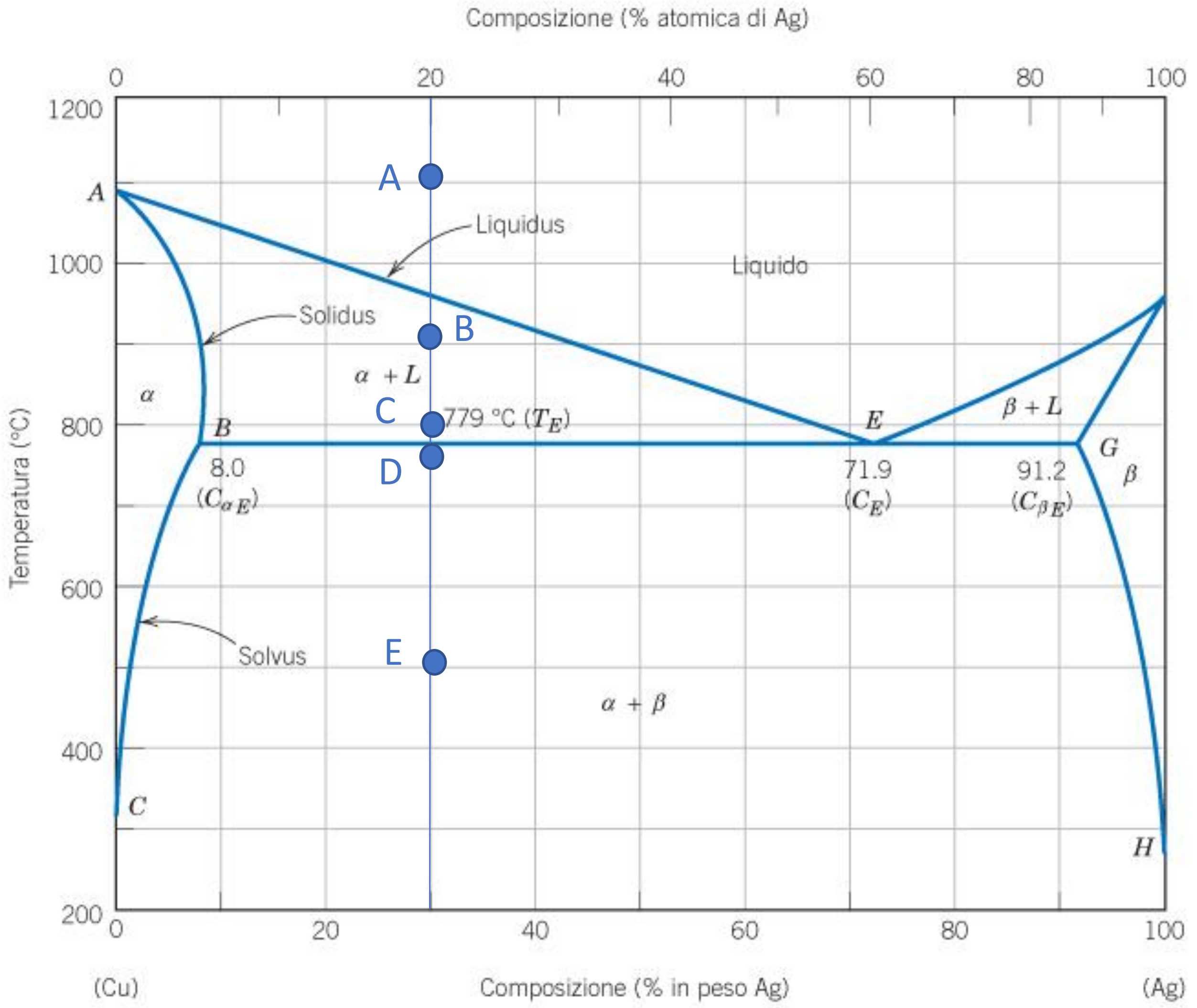
Exercise 4: describe the solidification of alloys with compositions x_B', x_B'', x_B''' ans x_B^{eut} by drawing schematic microstructures in significant points.



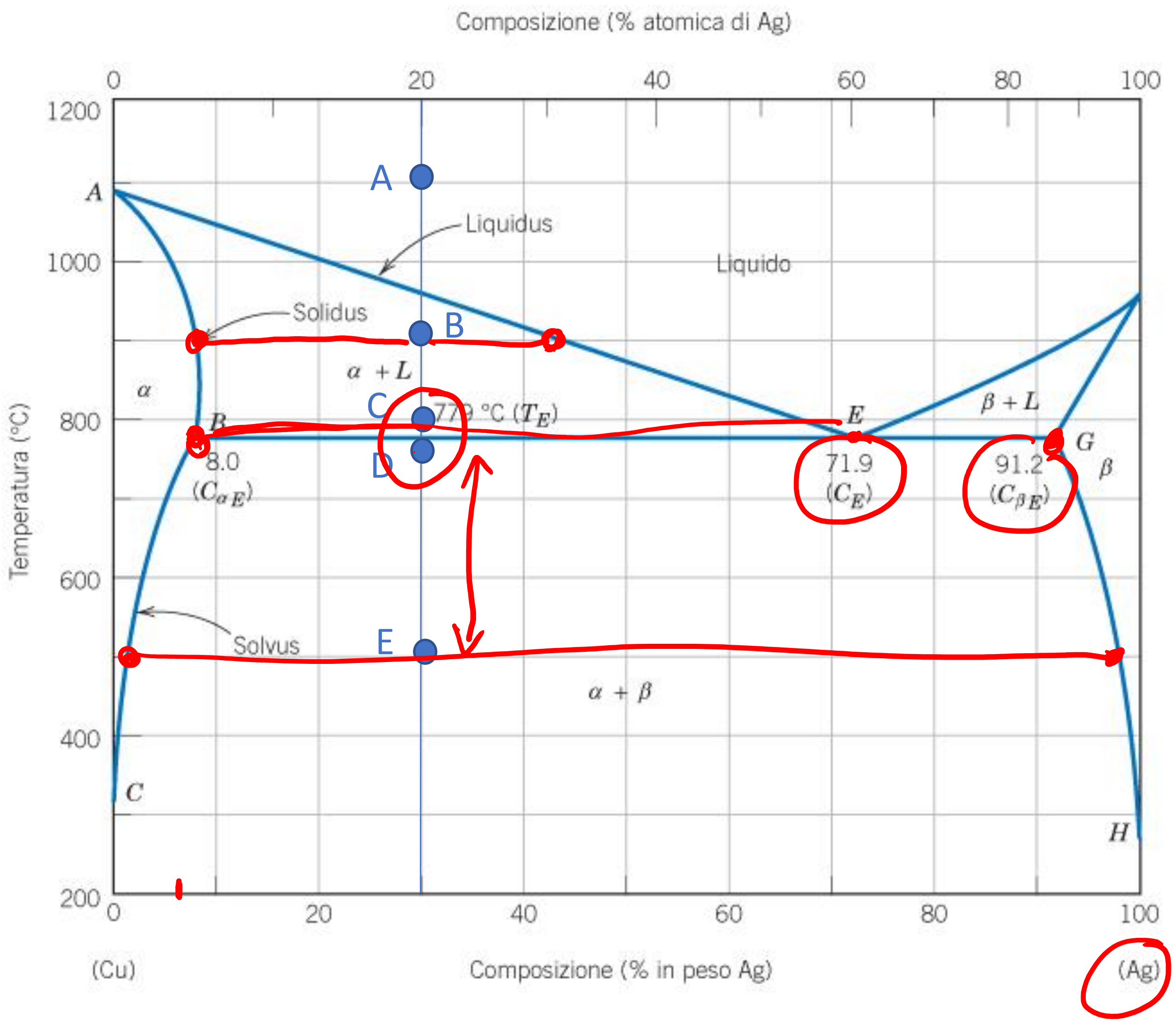


L(x_B')
L(x_B'')
L(x_B''')
L(x_B^eut)
L
S=α(x_B^α(T))
L(x_B^L(T))
S=α(x_B^α(T))
L(x_B^L(T))
S=α(x_B^α(T)=primary)
L(x_B^L(T))
α+L
S=α(x_B')
α
S=α(x_B')
S=α(x_B^α(T))
S=β(x_B^β(T))
S=(x_B^α(primary))
S=α(x_B^α,eut (T_eut))
S=β(x_B^β,eut (T_eut))
α+β
S=β(x_B^β,eut (T_eut))
S=α(x_B^α,eut (T_eut))
temperature
A
x_B'
x_B''
x_B'''
x_B^eut
B content

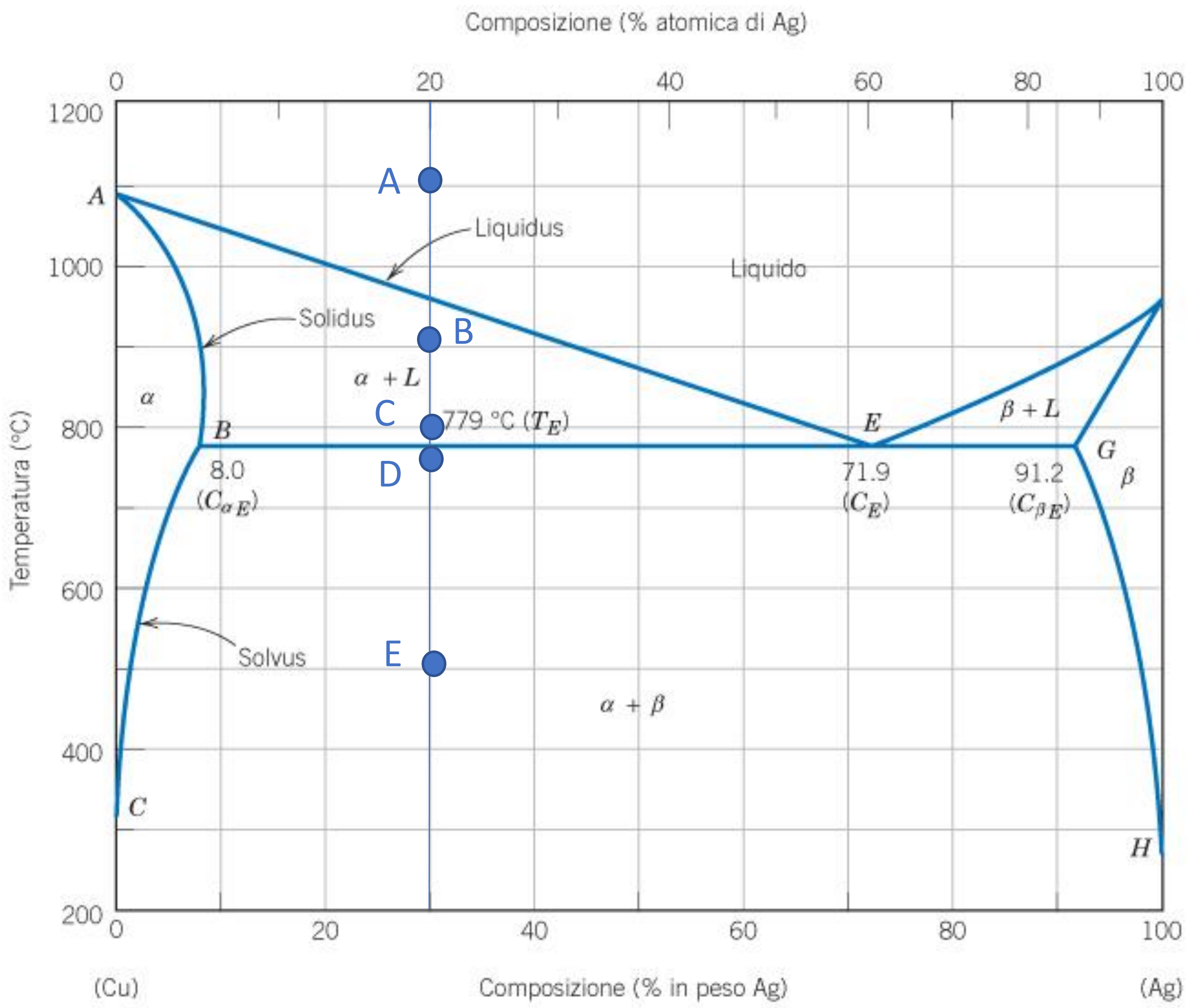
Exercise 5: determine the phase present, their composition and relative amounts in the points A, B, C (at the beginning of the eutectic transformation), D (at the end of the eutectic transformation) and E indicated in the phase diagram. Schematize the microstructure in each point.



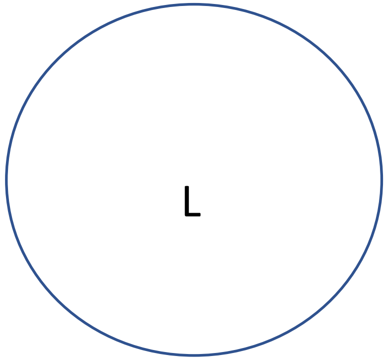
Exercise 5: determine the phase present, their composition and relative amounts in the points A, B, C (at the beginning of the eutectic transformation), D (at the end of the eutectic transformation) and E indicated in the phase diagram. Schematize the microstructure in each point.



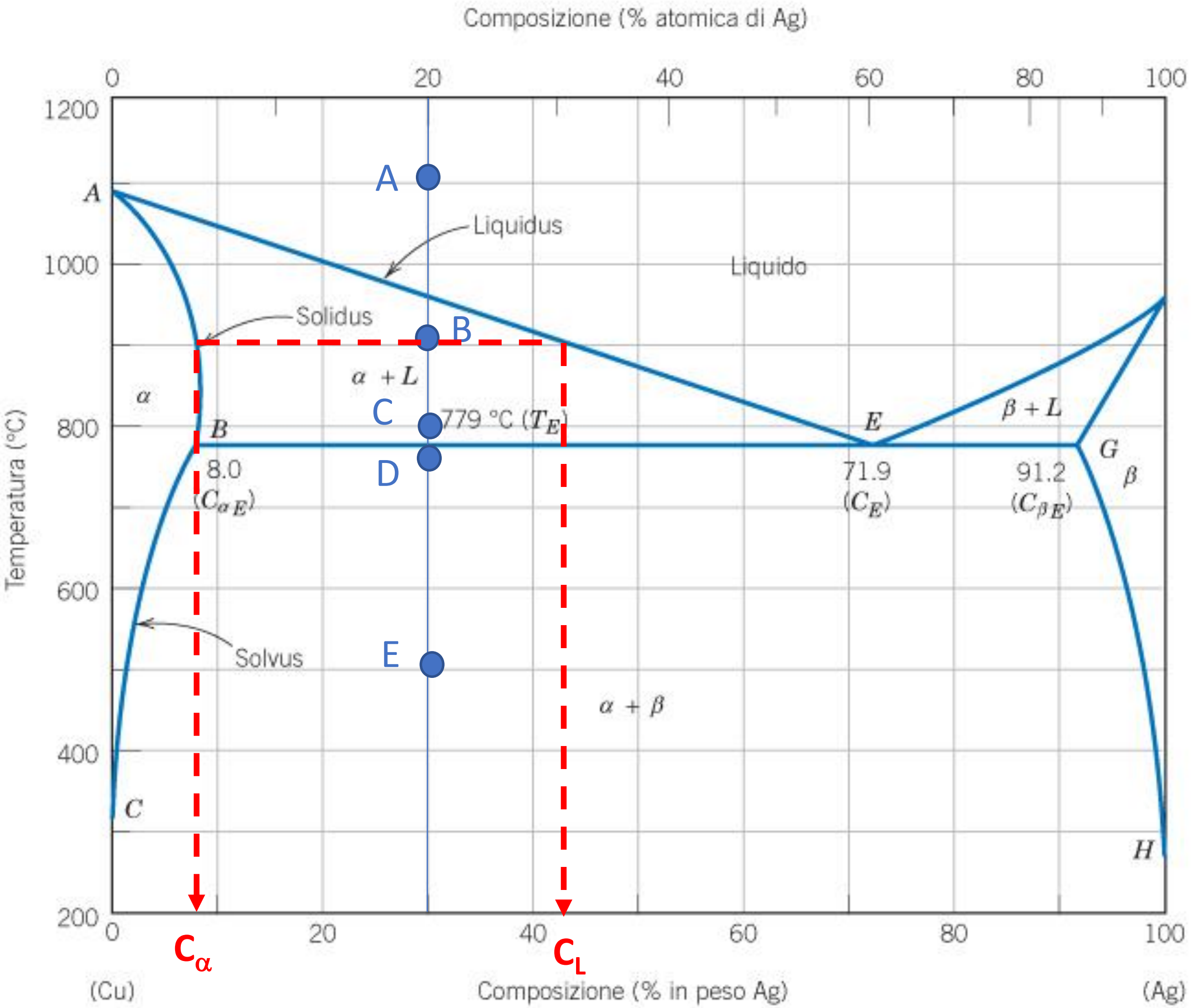
8% Ag

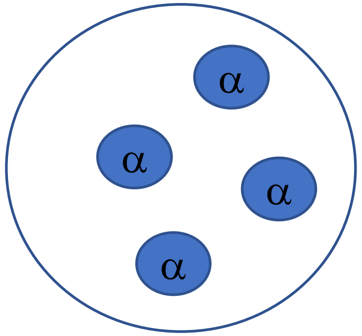


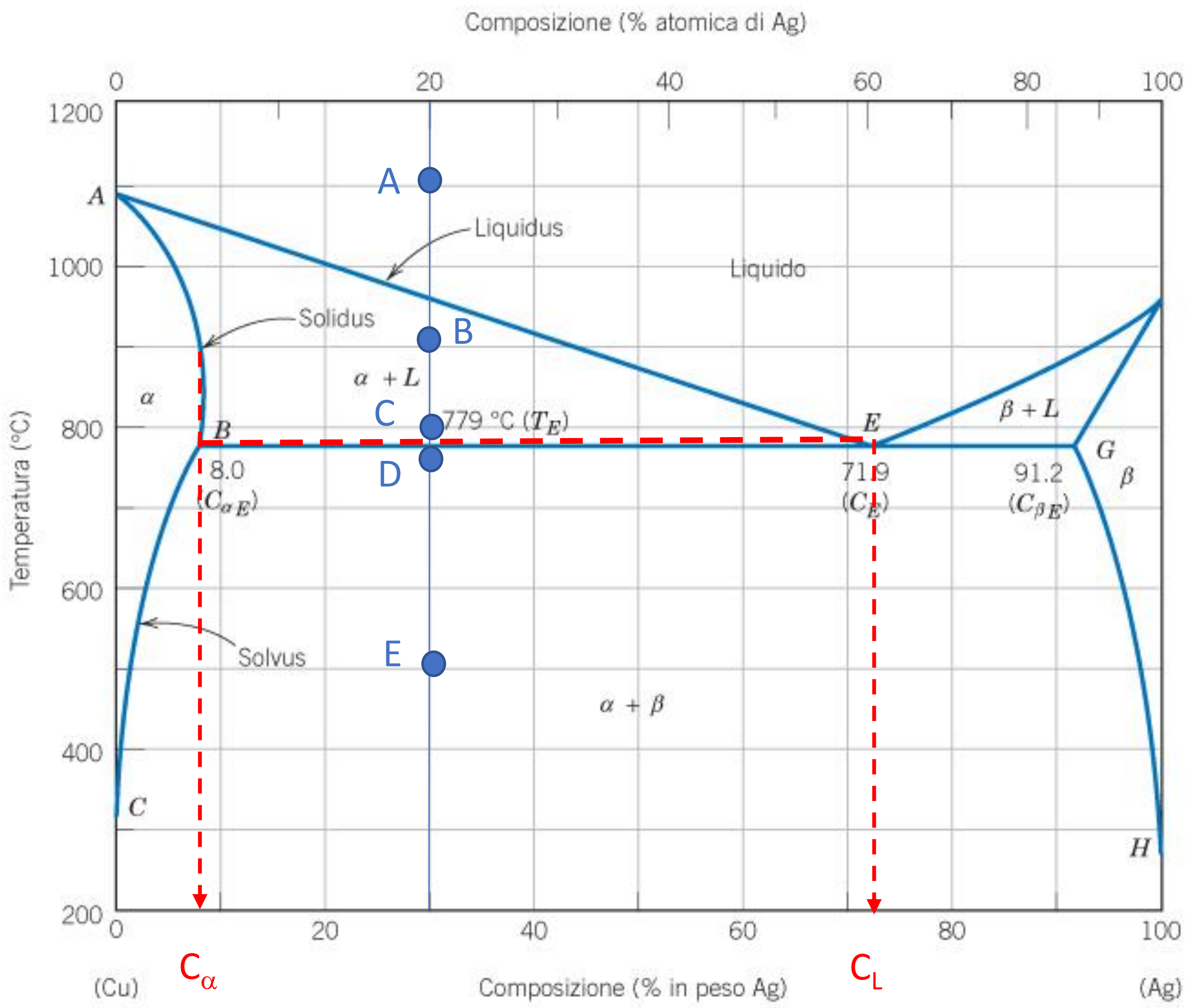
Composizione (% atomica di Ag)
0
20
40
60
80
100
1200
1000
800
600
400
200
Temperatura (°C)
A
B
C
D
E
Liquidus
Liquido
Solidus
α
α + L
779 °C (T_E)
8.0
($C_{\alpha E}$)
71.9
(C_E)
91.2
($C_{\beta E}$)
β + L
G
β
Solvus
α + β
H
(Cu)
Composizione (% in peso Ag)
(Ag)



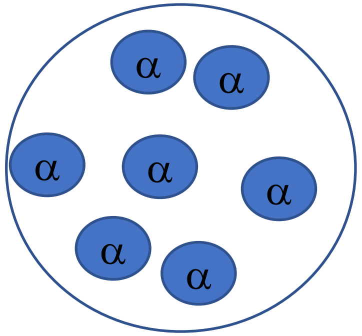
L


Composizione (% atomica di Ag)
0
20
40
60
80
100
1200
1000
800
600
400
200
Temperatura (°C)
A
B
C
D
E
Liquidus
Solidus
Liquido
α
α + L
779 °C (T_E)
B
8.0
($C_{\alpha E}$)
E
71.9
(C_E)
β + L
G
β
91.2
($C_{\beta E}$)
Solvus
α + β
C
H
0
20
40
60
80
100
C_α
C_L
(Cu)
Composizione (% in peso Ag)
(Ag)

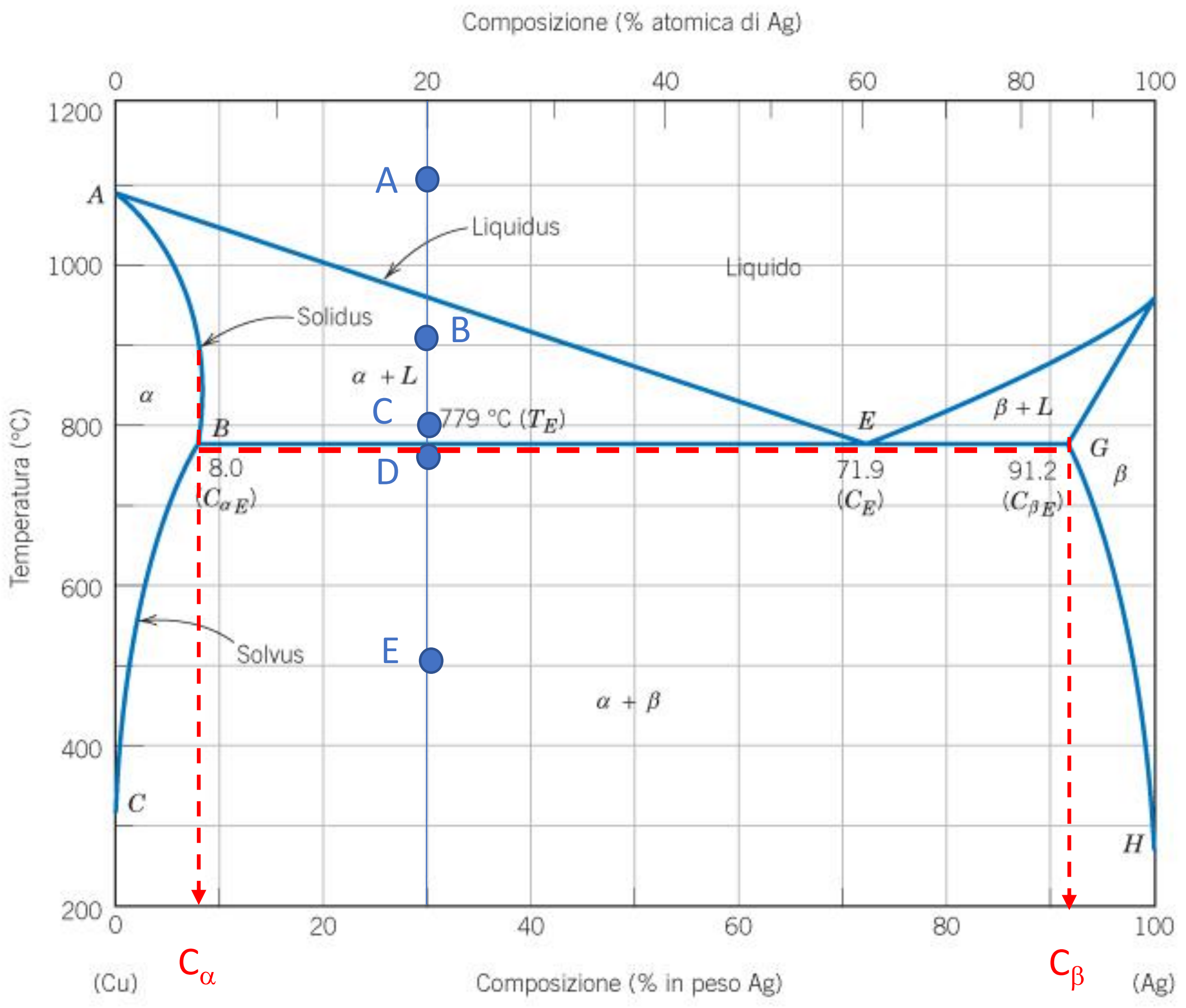

α
α
α
α



Composizione (% atomica di Ag)
0
20
40
60
80
100
1200
1000
800
600
400
200
A
Liquidus
Liquido
Solidus
B
α + L
α
C
779 °C (T_E)
B
E
β + L
G
β
8.0
($C_{\alpha E}$)
D
71.9
(C_E)
91.2
($C_{\beta E}$)
Temperatura (°C)
Solvus
E
α + β
C
H
0
20
40
60
80
100
C_α
C_L
(Cu)
Composizione (% in peso Ag)
(Ag)



α
α
α
α
α
α
α



Composizione (% atomica di Ag)
0
20
40
60
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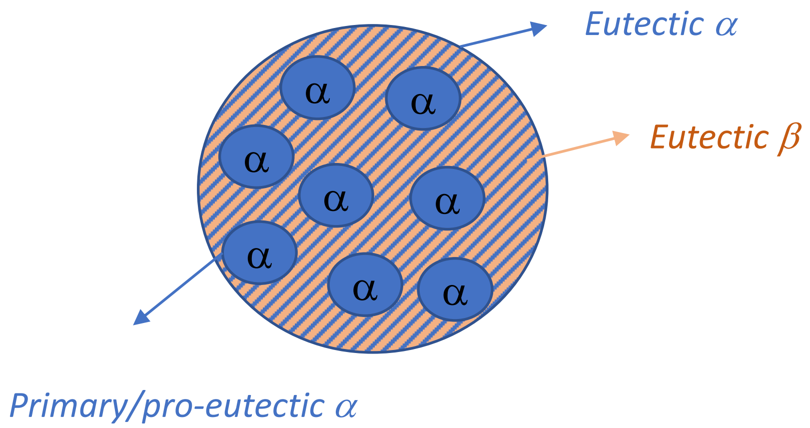
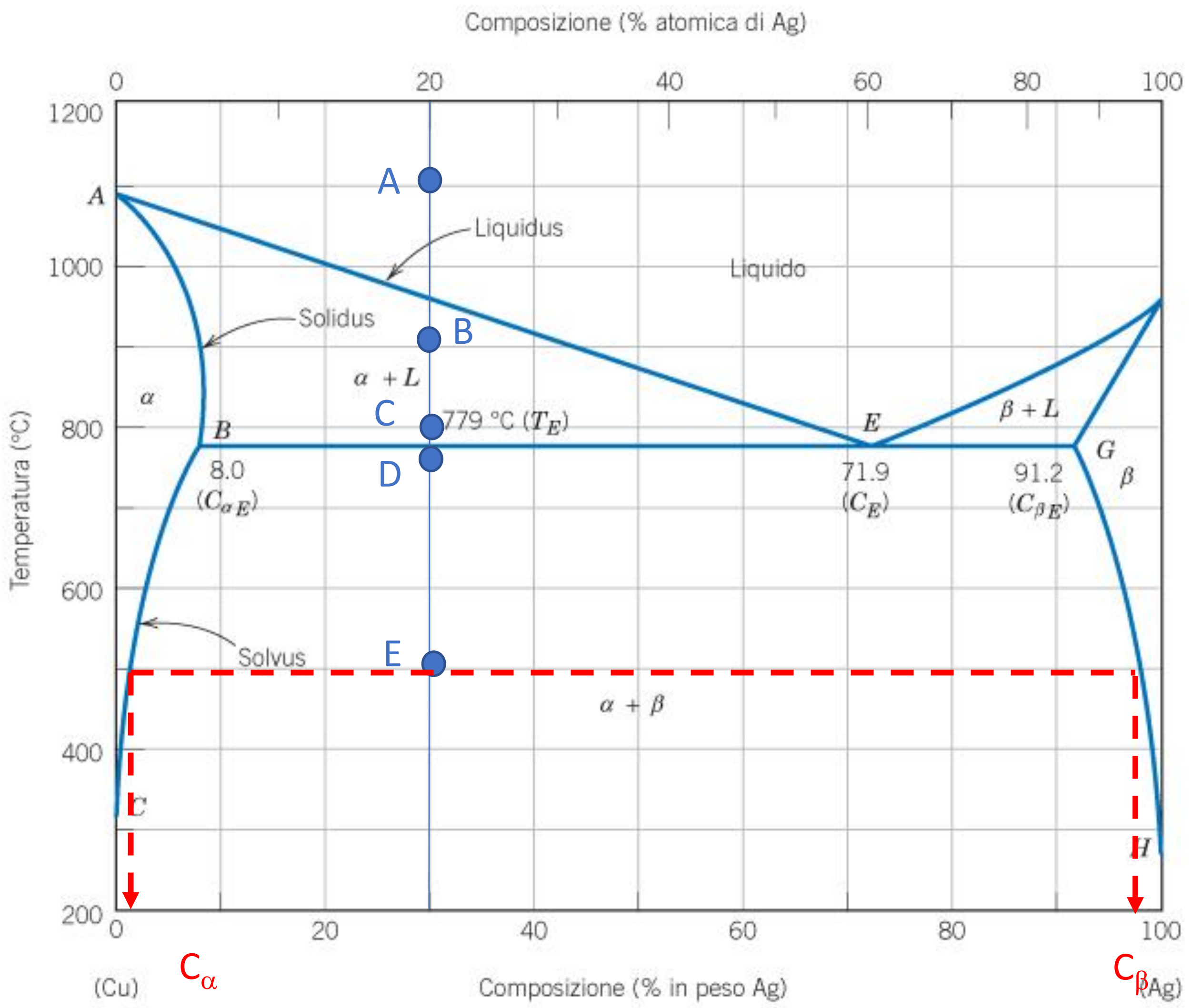
1200
1000
800
600
400
200
Temperatura (°C)
A
Liquidus
Liquido
Solidus
B
α + L
α
C
779 °C (T_E)
B
E
β + L
G
β
D
8.0
($C_{\alpha E}$)
71.9
(C_E)
91.2
($C_{\beta E}$)
Solvus
E
α + β
C
H
0
20
40
60
80
100
(Cu)
C_α
Composizione (% in peso Ag)
C_β
(Ag)



Eutectic α
Eutectic β
α
α
α
α
α
α
α
α
Primary/pro-eutectic α


Composizione (% atomica di Ag)
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20
40
60
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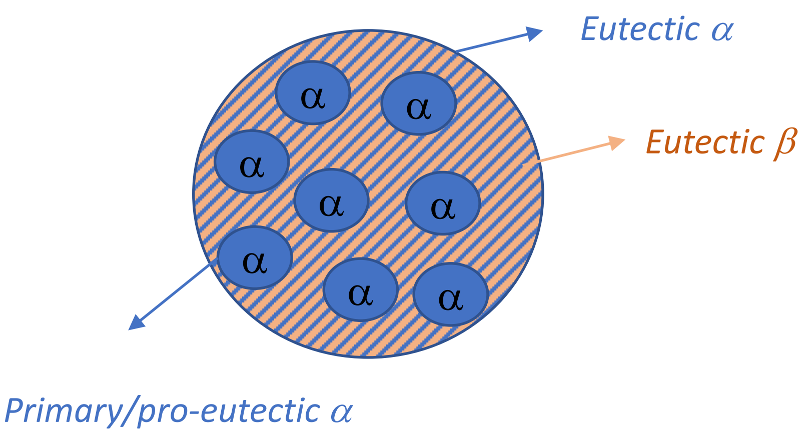
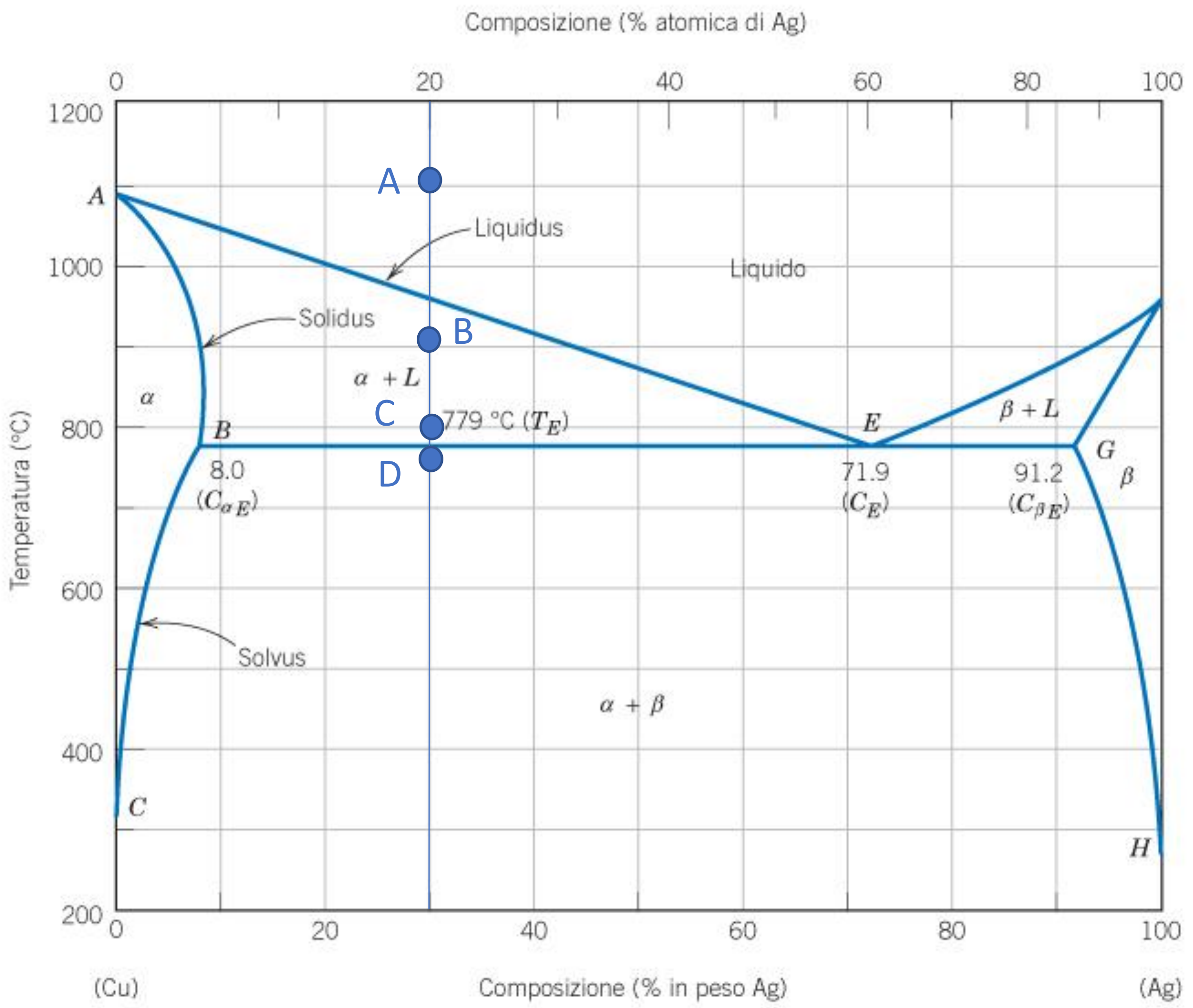
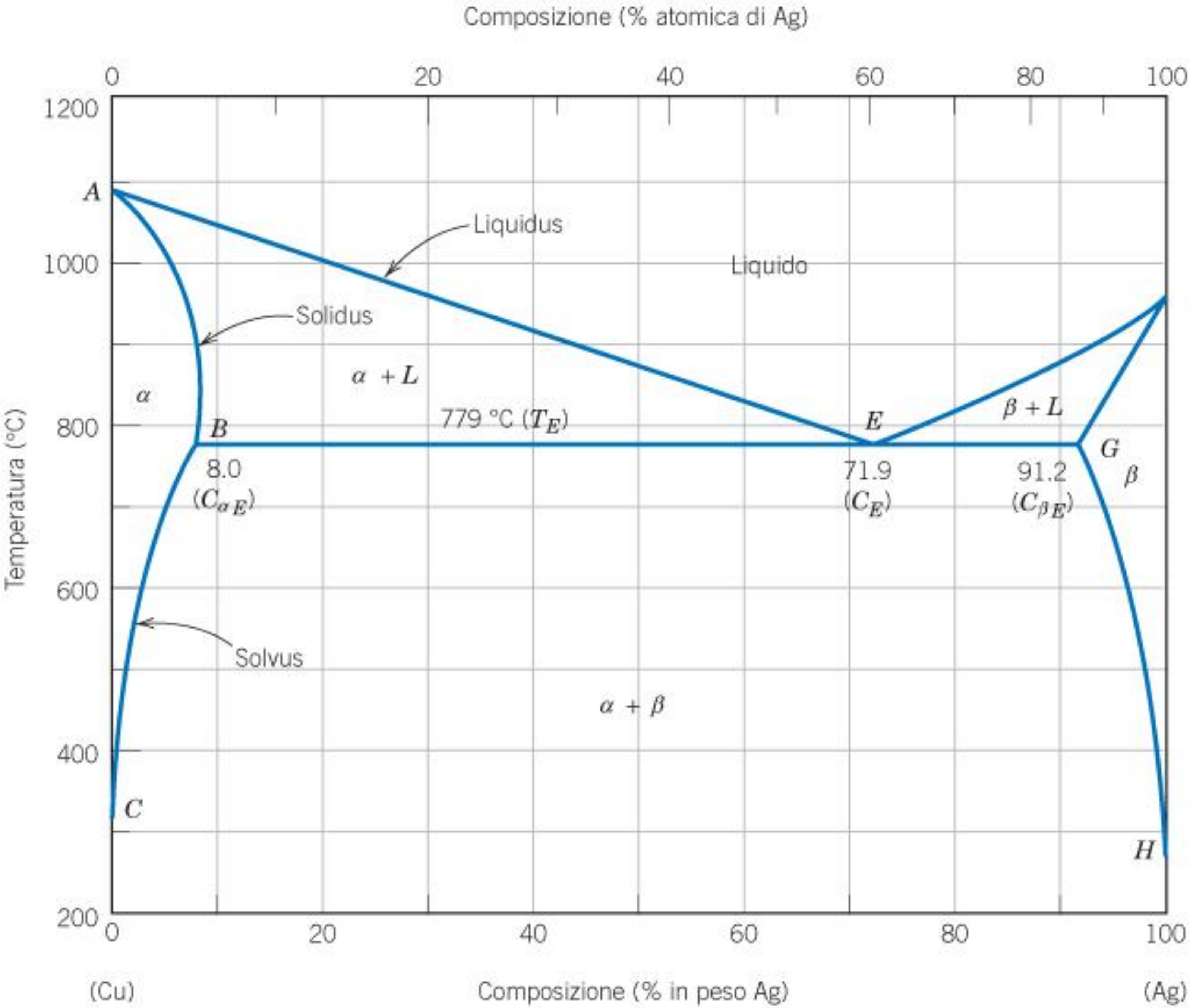
100
1200
1000
800
600
400
200
Temperatura (°C)
A
Liquidus
Liquido
Solidus
B
α + L
α
C
779 °C (T_E)
B
E
β + L
G
β
D
8.0
($C_{\alpha E}$)
71.9
(C_E)
91.2
($C_{\beta E}$)
Solvus
E
α + β
C
H
0
20
40
60
80
100
(Cu)
C_α
Composizione (% in peso Ag)
C_β
(Ag)


Eutectic α
Eutectic β
α
α
α
α
α
α
α
α
Primary/pro-eutectic α



Composizione (% atomica di Ag)
0
20
40
60
80
100
1200
1000
800
600
400
200
Temperatura (°C)
A
B
C
D
Liquidus
Liquido
Solidus
α + L
α
779 °C (T_E)
8.0
($C_{\alpha E}$)
E
71.9
(C_E)
β + L
G
91.2
($C_{\beta E}$)
β
Solvus
α + β
C
H
0
20
40
60
80
100
(Cu)
Composizione (% in peso Ag)
(Ag)


Composizione (% atomica di Ag)
0
20
40
60
80
100
1200
1000
800
600
400
200
Temperatura (°C)
A
Liquidus
Liquido
Solidus
α
α + L
779 °C (T_E)
B
E
β + L
G
β
8.0
($C_{\alpha E}$)
71.9
(C_E)
91.2
($C_{\beta E}$)
Solvus
α + β
C
H
0
20
40
60
80
100
(Cu)
Composizione (% in peso Ag)
(Ag)

Eutectic phase diagram